USD/CRC · MONEX — the calendar strategy: dynamics, rule logic & VWAP-priced results

2,663 sessions · 2014-12-08 → 2026-06-19 · long/short USD · $1,000,000 per trade · **priced VWAP-to-VWAP**, net of 0.65 CRC round-trip · no technical indicators

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The recommended strategy is a **calendar (quincena) rule with a slow-volume size trim**: short USD into Costa Rica's mid-month IVA / payroll deadline (within 6 business days of it), long the rest of the month, at **half size whenever a slow 20-day volume regime disagrees** with the trade. Selection is disciplined — the variants are ranked **in-sample** (first 60% of history) and every headline figure here is the **out-of-sample** realisation on the 40% held out after **2021-11-18**, so nothing you read below was tuned on the data it is measured on. On that held-out window the rule earns **$123k/yr at Sharpe 3.55** (full-sample Sharpe 3.31, ~18 trades/yr) — and it posts **the family's best out-of-sample Sharpe, strengthening from 3.22 in-sample to 3.55 out** where the strong flat-sized entries give the edge back, because the slow-vol trim cuts the fat losing trades (out-of-sample per-trade skew **+0.74**, worst trade $-18k vs $-50k for the crude base rule). An optional **dynamic trailing-stop exit** (A5) is a second, independent way to cut the same reversion tail. Every number below is VWAP-priced; secondary signals are kept in tabs but default to this rule.

3.55
out-of-sample Sharpe
recommended rule · $123k/yr · 40% held out

3.22 → 3.55
in-sample → out-of-sample
family-best OOS Sharpe — it strengthens where the flat rules decay

3.31
Sharpe (full sample)
~18 trades/yr · $1M/trade · VWAP-priced

+0.74
OOS per-trade skew
slow-vol trim halves the worst trade to $-18k

3.74
Sharpe with exit overlay
optional trailing stop cuts DD to $-26k (A5)

29 vs 18 M
volume: colón up vs down
USD sellers trade in size

One accounting basis throughout. Every figure in this report — every part, every table — is priced the same way: **VWAP-to-VWAP** next session, **$1,000,000** per trade, net of the **0.65 CRC** round-trip slippage (0.325/side), and annualised on the **231** sessions/yr MONEX actually trades — not the 252 equity-market convention (which would overstate $/yr by ~9% and Sharpe by ~4.5%). All modules read this basis from a single source (`src/basis.py`), so the per-module tables (Parts A–F) and the unified ranking (Part R) now agree by construction. The **unified ranking (Part R)** remains the master cross-strategy comparison.

PART A · THE RECOMMENDED STRATEGY — CALENDAR / QUINCENA RULE

A1. The rule & the numbers ($1M per trade, VWAP-priced, net of cost)

Built up one design choice at a time. **Sharpe (full)** is the whole history; **Sharpe (OOS)** is the held-out 40% after 2021-11-18 — the honest number. The winner is the one that is best *out of sample*, not the one with the biggest headline P&L.

VERSION	PER YEAR (FULL)	SHARPE (FULL)	SHARPE (OOS)	TRADES/YR	MAX DD	NOTE
Base quincena (short if day ≤ 15)	$80k	1.97	2.08	22.2	$-62k	crude ancestor; left-skewed tail
Real calendar (≤6 bd), flat size	$114k	2.88	2.79	22.2	$-48k	deadline-anchored entry; OOS slips vs IS
Refined (fixed 5–15) + slow-vol	$94k	3.23	3.35	18.2	$-37k	slow-vol holds OOS, fixed window
Calendar + slow-vol RECOMMENDED · BEST OOS	$95k	3.31	3.55	18	$-37k	deadline anchor + tail-cutting trim

Exact logic (the trading calendar)

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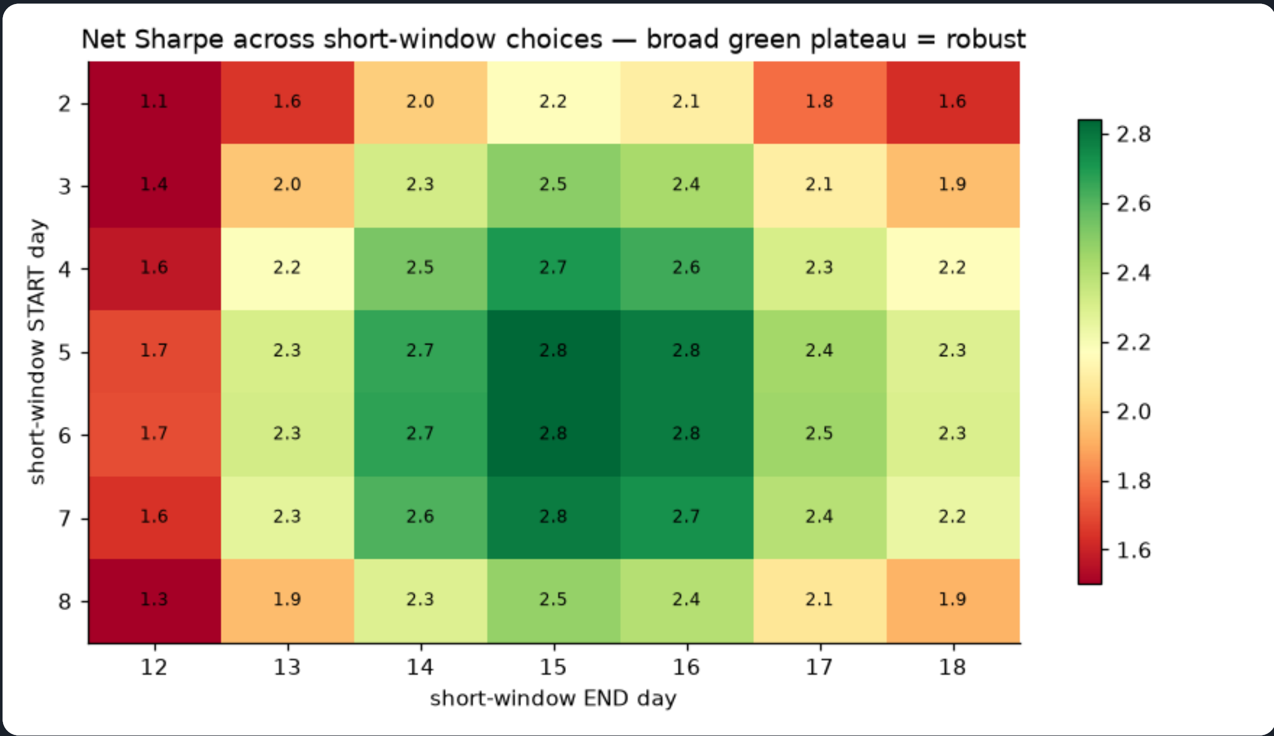
≤ 6 business days before the IVA/quincena deadline (through it) → SHORT USD
   (firms sell USD to raise colones for the 15th tax + payroll → colón
strengthens)
otherwise → LONG USD (supply fades, USD drifts up)
size trim: HALF size whenever a slow 20-day volume regime disagrees with the
trade
```

Two design choices, each justified before looking at the test window. **Entry** anchors the short window to Costa Rica's actual IVA (D-104) / mid-month quincena deadline — the 15th, rolled to the next business day — so it shifts with weekends and holidays (the fixed day-of-month 5–15 window is its holiday-blind twin, a near-identical trade population). **Sizing** trims to half whenever a slow 20-day volume regime disagrees, cutting the biggest losing trades without changing direction or adding turnover. In-sample the two slow-vol variants are a statistical tie (Sharpe 3.22 vs 3.3, well inside noise for ~250 trades); we keep the calendar anchor for operational robustness, and out of sample it is the stronger of the two (Sharpe 3.55 vs 3.35).

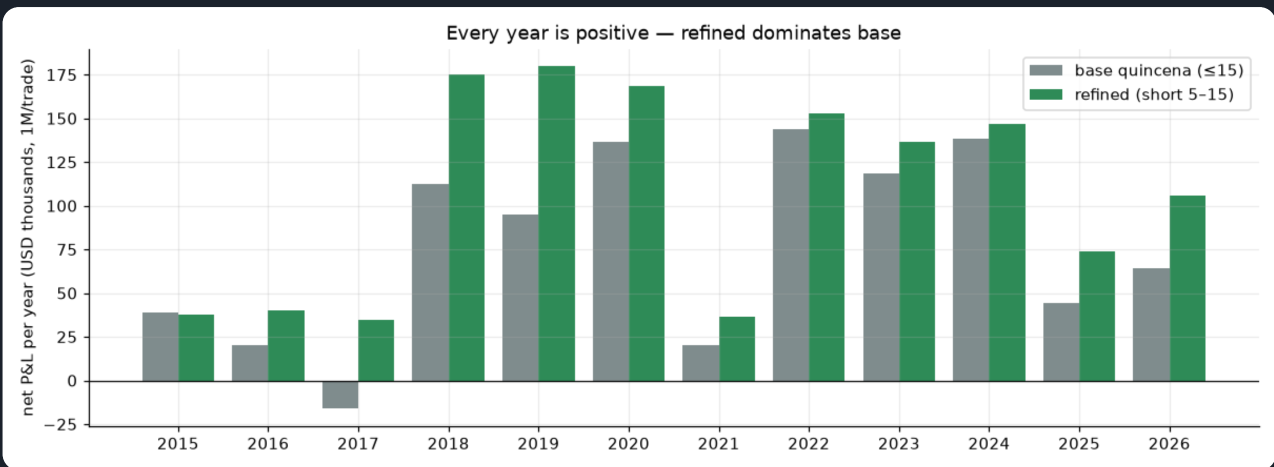
A2. Is it overfit? Window sensitivity & every-year P&L

Net Sharpe across every choice of the short-window start & end



The result is a broad green plateau (Sharpe 1.05–2.84), not a lucky point — days 5→15 is simply the natural mid-month supply window, and neighbours all work.

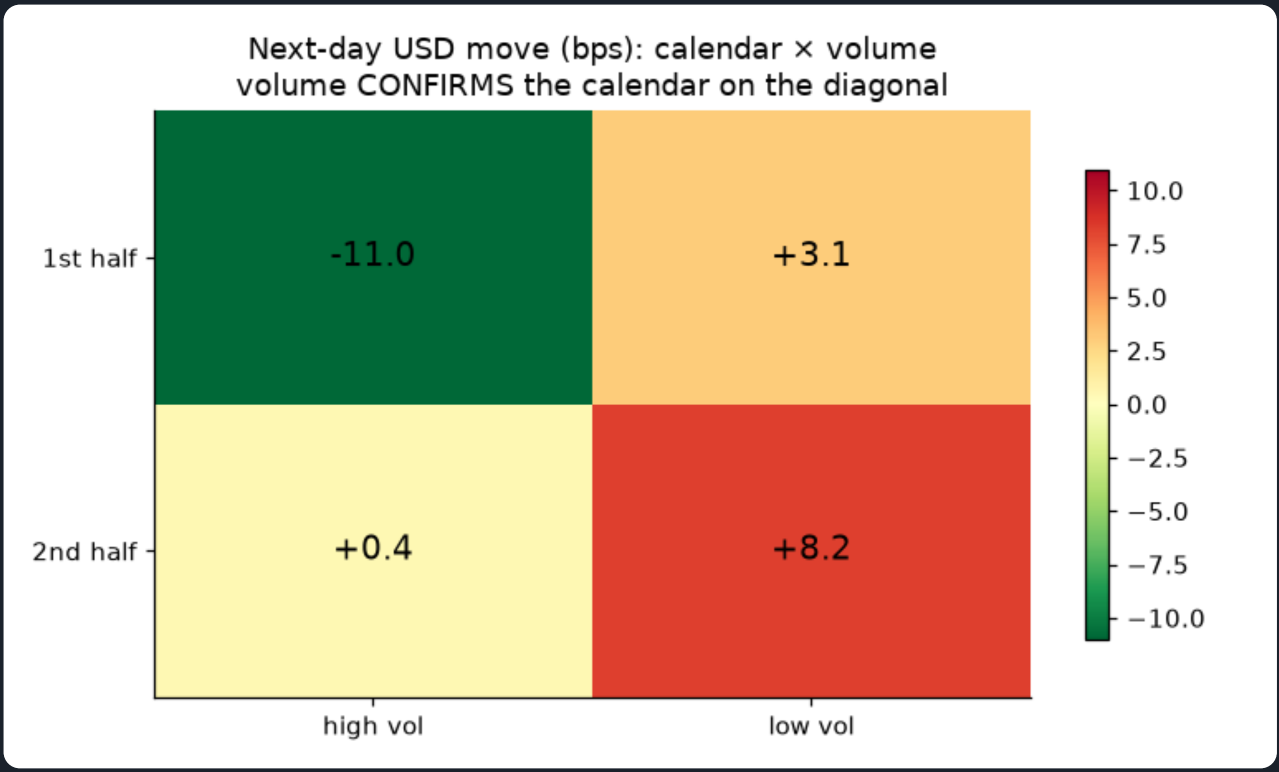
Net P&L by year — base vs refined ($1M/trade, VWAP-priced)



Positive in all 12 calendar years (worst $34k), including the 2015–17 pegged years where the volume signal was dead. The refined version dominates the base almost every year.

A3. How volume fits in — confirmation, not a daily signal

Next-day USD move: calendar × volume (bps)



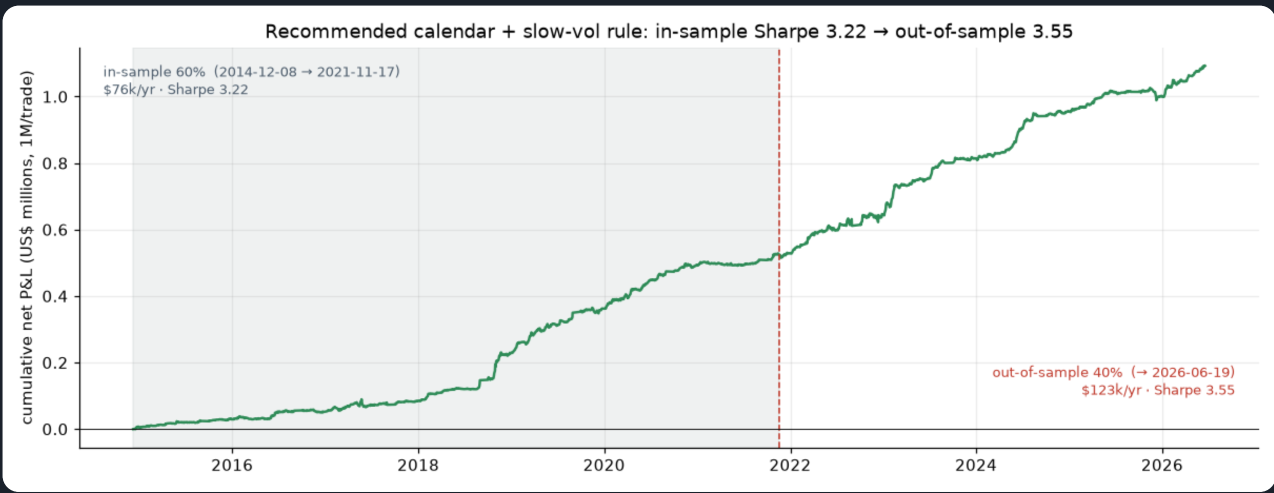
Volume CONFIRMS the calendar on the diagonal: first-half + high volume is strongly USD-down (sell USD), second-half + low volume strongly USD-up (buy USD). But trading volume day-by-day adds turnover that costs more than it adds — so we fold it in as a SLOW regime filter that only trims size, which is what lifts the Sharpe to 3.23 and cuts the drawdown to $-37k.

A4. Sample vs out-of-sample — how the whole family is selected

> This is the discipline behind the recommendation. The rules carry **no fitted parameters** (the 20-day volume window, the ≤6-day anchor and the half-size trim are fixed a-priori choices, not optimised), but we still split history chronologically 60/40 at **2021-11-18**, **rank on the in-sample Sharpe**, and then read the out-of-sample column as the honest confirmation. The pattern is decisive: **the slow-vol size trim is the design choice that matters** — the strong flat-sized entries give the edge back out of sample (the real-calendar entry 3.14→2.79, the fixed window 3.19→2.67), while both slow-vol variants hold or improve (▲). The recommended **calendar + slow-vol** posts the family's best out-of-sample Sharpe, 3.55 — up from 3.22 in sample.

RULE (CHRONOLOGICAL 60/40 SPLIT AT 2021-11-18, NO RE-FITTING)	IS / YR	IS SHARPE	OOS / YR	OOS SHARPE	OOS VS IS
Base quincena (≤15)	$62k	1.96	$106k	2.08	▲
Real calendar (≤6 bd), flat size	$97k	3.14	$140k	2.79	▼
Refined (fixed 5–15) + slow-vol	$77k	3.3	$120k	3.35	▲
Calendar + slow-vol RECOMMENDED	$76k	3.22	$123k	3.55	▲

Recommended calendar + slow-vol rule (VWAP-priced): in-sample vs out-of-sample equity



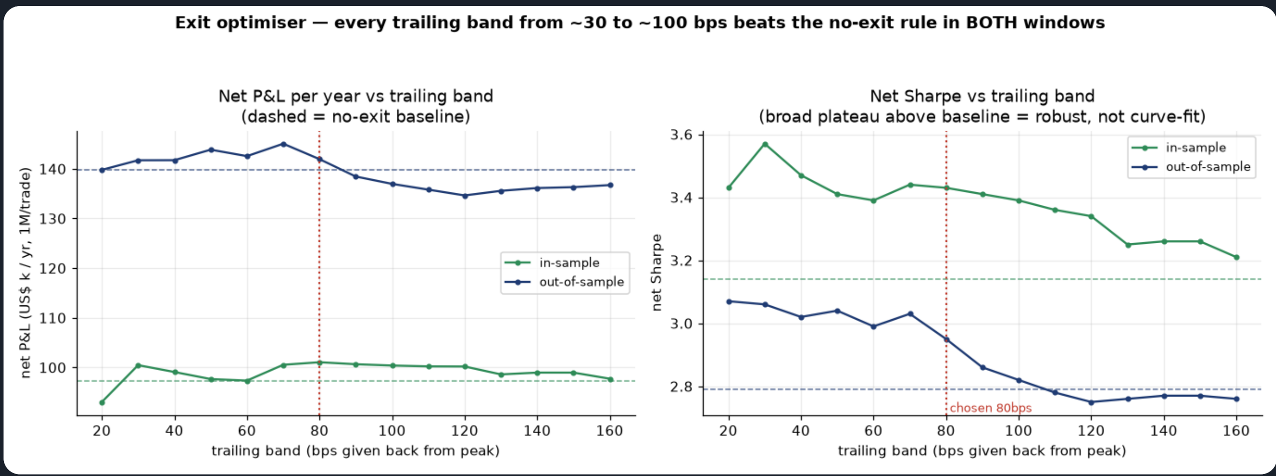
In-sample 2014-12-08 → 2021-11-17 (Sharpe 3.22) and out-of-sample → 2026-06-19 (Sharpe 3.55, $123k/yr). The out-of-sample slope is if anything steeper — the post-2021 float is where the cash-flow cycle is most tradeable, and the slow-vol trim keeps the held-out drawdowns shallow.

A5. An alternative tail control — a dynamic exit (stop-loss / take-profit)

> The recommended rule already cuts the reversion tail through **sizing** (the slow-vol trim). A dynamic **trailing stop** is a second, independent way to do it through **timing**, shown here on the flat-sized calendar entry so the mechanism is isolated. Because the edge **front-loads** — the colón strengthens *into* the deadline and reverts after — a trade often peaks mid-hold and gives the move back; once the running P&L gives back 80 bps from its best level we bank it and stay flat until the calendar opens the next trade. This **never changes the entry logic** — it can only remove the losing tail. Both parameters are optimised on the **in-sample 60% only**; the out-of-sample 40% below is untouched. Treat it as an optional overlay: it and the slow-vol trim are two routes to the same tail control, not a stack to apply blindly together.

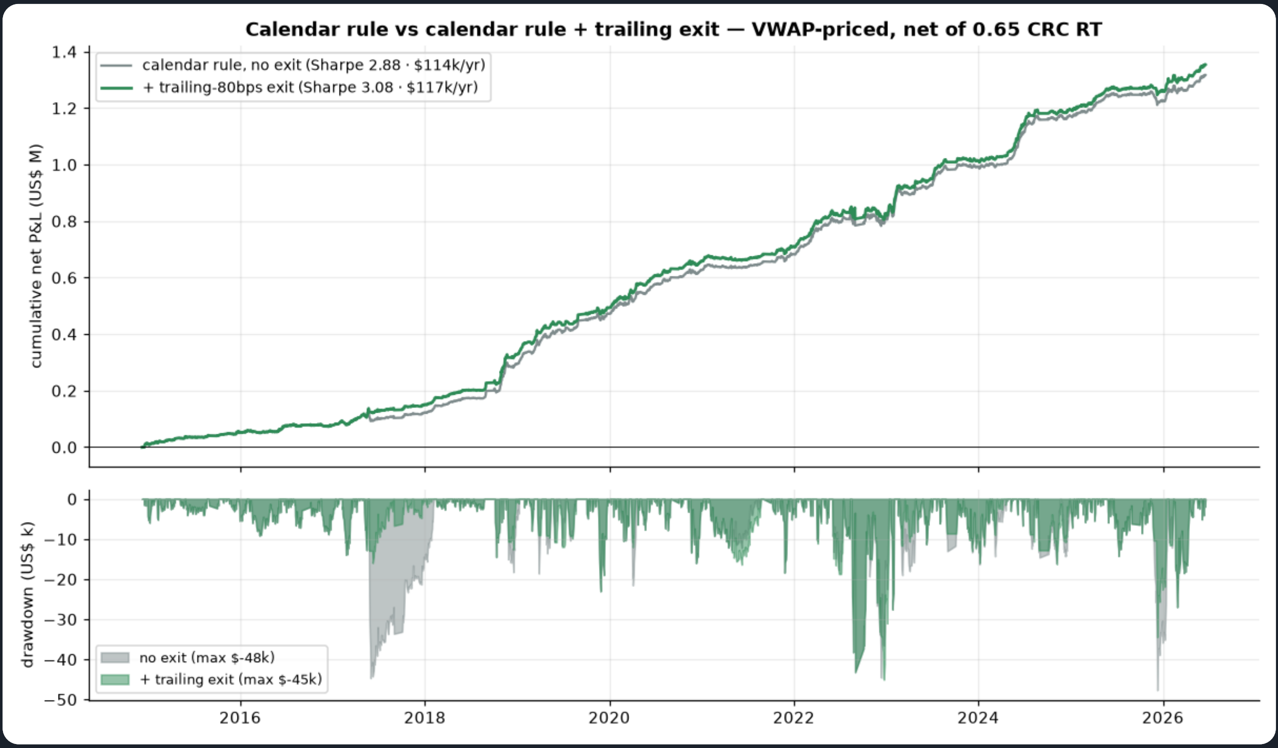
CALENDAR RULE · $1M/TRADE · VWAP-PRICED, NET COST	FULL / YR	SHARPE	MAX DD	OOS / YR	OOS SHARPE
No exit — always in the market the rule from A1–A4	$114k	2.88	$-48k	$140k	2.79
+ trailing-stop exit (80 bps) RECOMMENDED tuned on in-sample P&L; biggest raw P&L	$117k	3.08	$-45k	$142k	2.95
+ trailing (30) & hard floor (60) bps tuned on in-sample Sharpe; best risk-adjusted	$116k	3.74	$-26k	$140k	3.92

Exit optimiser — net P&L and Sharpe across the trailing band



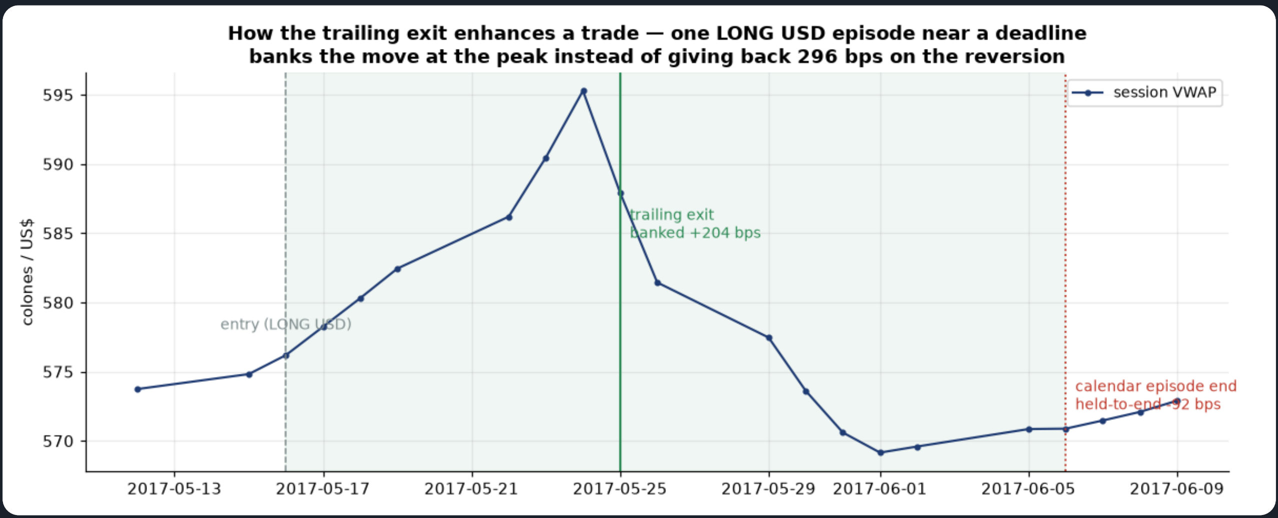
This is the search itself. Every trailing band from ~30 to ~100 bps beats the no-exit rule (dashed) in **both** the in-sample and out-of-sample windows on Sharpe, and lifts in-sample P&L across the whole range — a broad plateau, not a single lucky point (the same robustness test we applied to the calendar window in A2). The chosen 80 bps maximises in-sample P&L/yr.

Calendar rule vs calendar rule + trailing exit — equity & drawdown



Adding the exit lifts full-sample P&L from $114k/yr to $117k/yr (Sharpe 2.88 → 3.08) and — with the hard floor variant — cuts the worst drawdown from $-48k to $-26k. The green curve sits above the grey throughout, most visibly through the deep 2017–18 drawdown.

How the exit enhances one trade — bank the move at the peak, skip the reversion



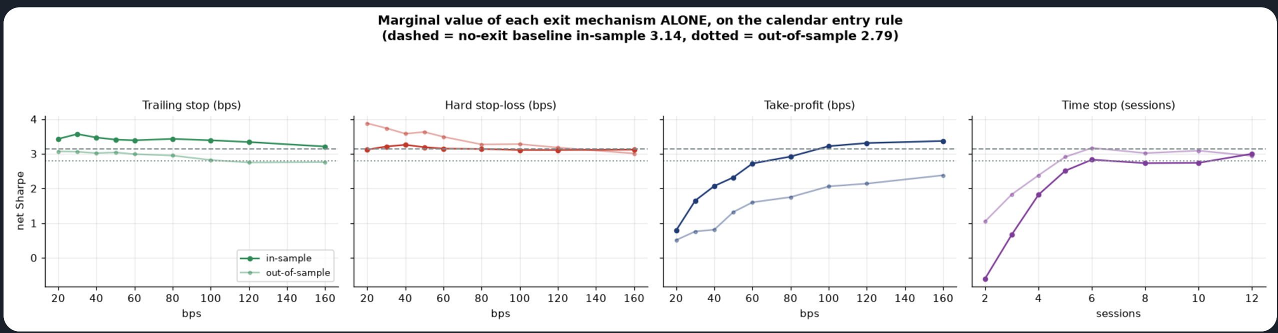
A concrete long-USD trade: the position banks +204 bps at the trailing exit near the peak, instead of holding to the calendar's episode end and watching the same trade turn into a -92 bps result on the post-deadline reversion — that one exit is worth ~296 bps. The overlay does this systematically across trades; it is the mechanism, on a single trade.

A5b. The full exit engine — which stop actually helps (newest run, 231-session basis)

> A5's trailing overlay was tuned by `exits.py`. The newest run (`exit_lab.py`) rebuilds it as a **four-mechanism engine** — trailing stop, hard stop-loss, take-profit and time stop — each tested **alone** before any are combined, everything selected on the in-sample 60% only and priced on the correct **231 sessions/yr**. The verdict is sharper than before: **a stop-loss helps, a take-profit actively hurts.** The recommended overlay is **trailing 30 bps + hard stop 40 bps, no take-profit** — in-sample Sharpe **3.7** → out-of-sample **3.92** (the overlay does not decay out of sample), cutting the worst drawdown from $-48k to $-26k.

MECHANISM (ALONE, ON THE CALENDAR ENTRY)	BEST SETTING	IS SHARPE	OOS SHARPE	VERDICT
Trailing stop	30 bps	3.57	3.06	helps
Hard stop-loss	40 bps	3.26	3.58	helps — more OOS than IS
Take-profit	160 bps	3.37	2.38	HURTS out-of-sample
Time stop	12 sessions	3.0	2.94	no value
Baseline — no exit (calendar entry only)	—	3.14	2.79	reference

Marginal value of each exit mechanism ALONE, on the calendar entry

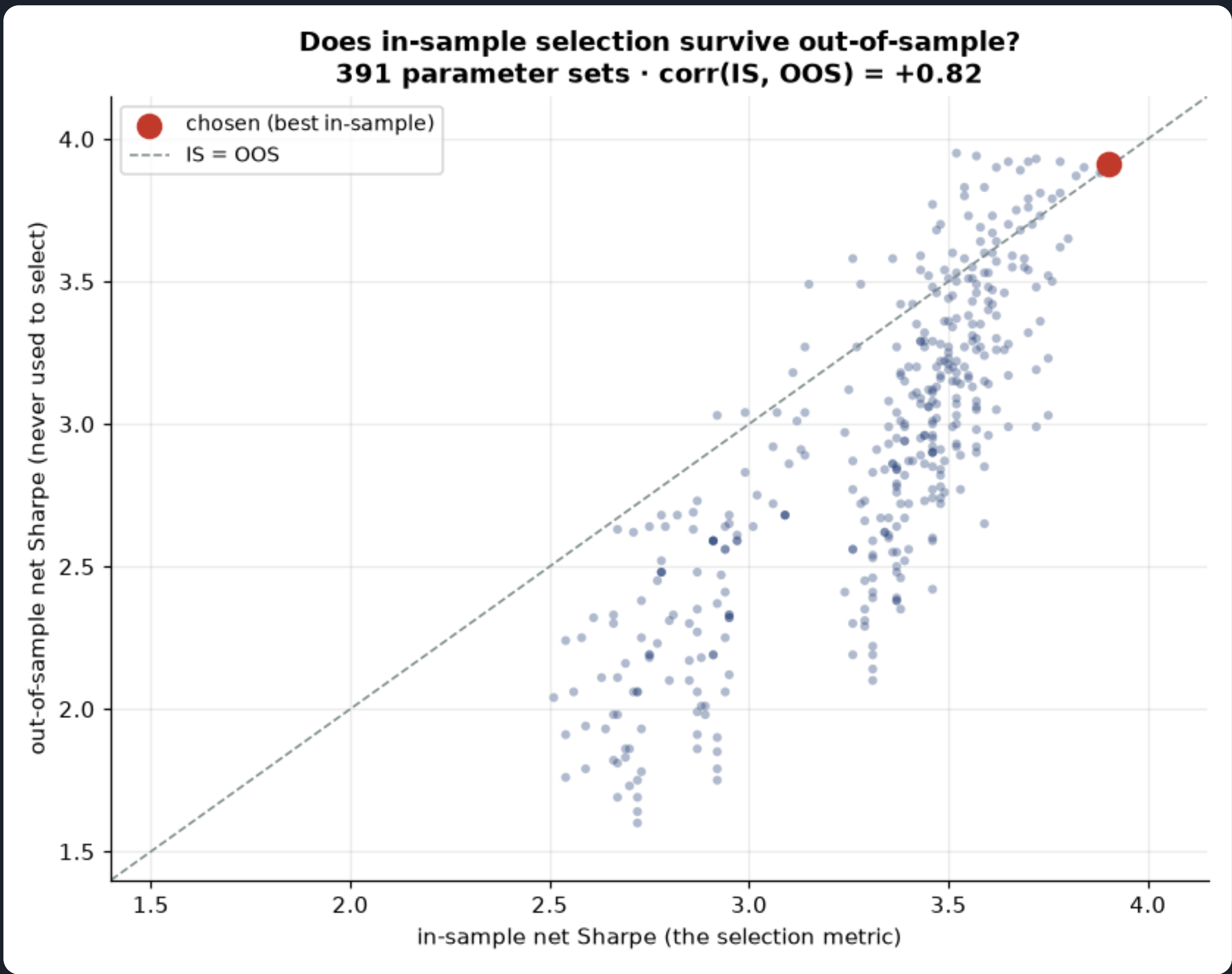


Each mechanism swept on its own (dashed = no-exit in-sample baseline 3.14, dotted = out-of-sample 2.79). Trailing and hard stops lift Sharpe across a broad band; the take-profit panel slopes the wrong way — tightening it costs Sharpe — and the time stop adds nothing.

> **Take-profit is the clear negative result, worth stating plainly.** Added on top of the trail 30 / floor 40 overlay, tightening the target degrades the out-of-sample Sharpe monotonically (3.92 with none → 3.04 at 80 bps). The calendar edge is a *drift* that accrues while the trade is held into the deadline, so capping the winner truncates exactly the move the strategy exists to capture.

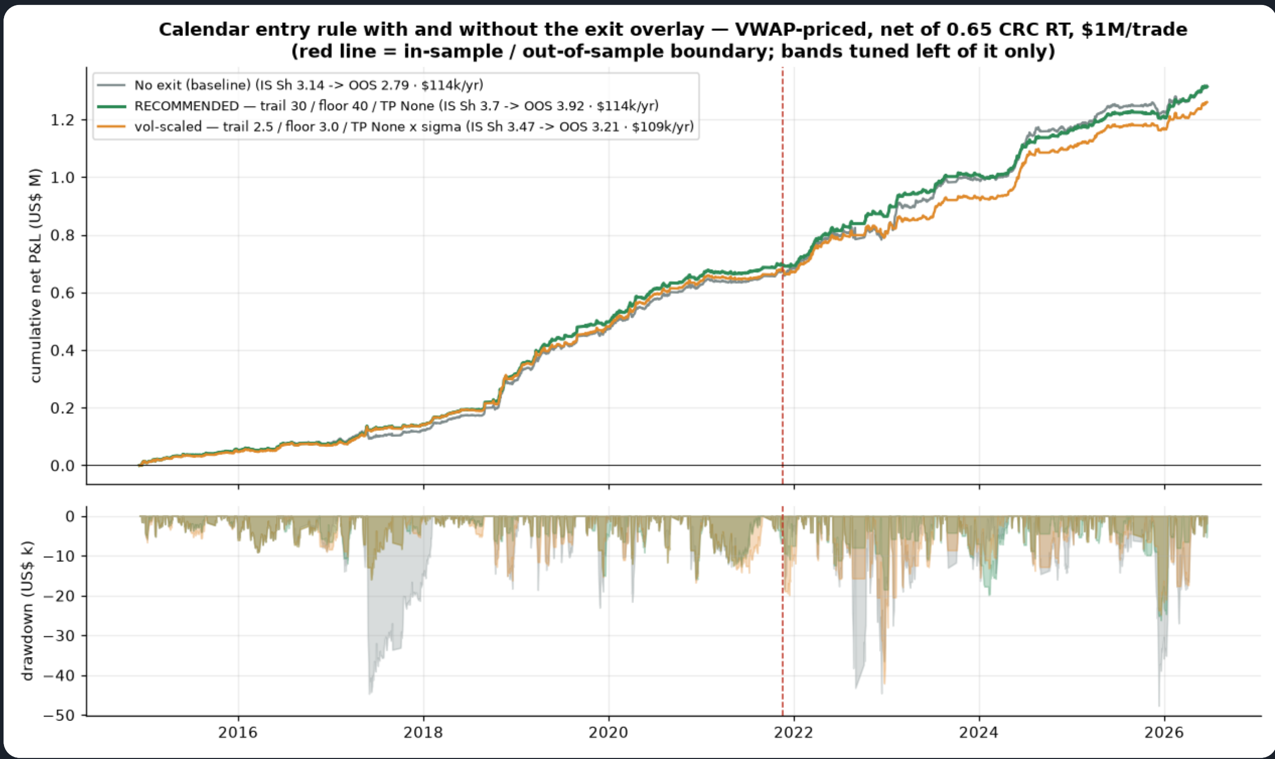
TAKE-PROFIT ADDED TO TRAIL 30 + FLOOR 40	IS SHARPE	OOS SHARPE	IS $/YR	OOS $/YR
none	3.7	3.92	$101k	$134k
320 bps	3.9	3.91	$104k	$131k
240 bps	3.88	3.88	$97k	$128k
160 bps	3.76	3.5	$89k	$109k
120 bps	3.64	3.26	$83k	$97k
80 bps	3.14	3.04	$64k	$82k

Does in-sample selection survive out-of-sample? Every parameter set plotted



All 391 joint-grid cells, in-sample Sharpe (the selection metric) vs out-of-sample. corr(IS, OOS) = +0.82 — in-sample ranking genuinely predicts out-of-sample for this parameter family. The chosen point (red) sits on the IS=OOS diagonal, not above it. Deflated-Sharpe P(true>0) ≈ 1.0 over the 391 trials; the 18 neighbours of the winner hold a median 3.73 (96% of peak) — a plateau, not a spike.

Calendar entry with and without the exit overlay — equity & drawdown

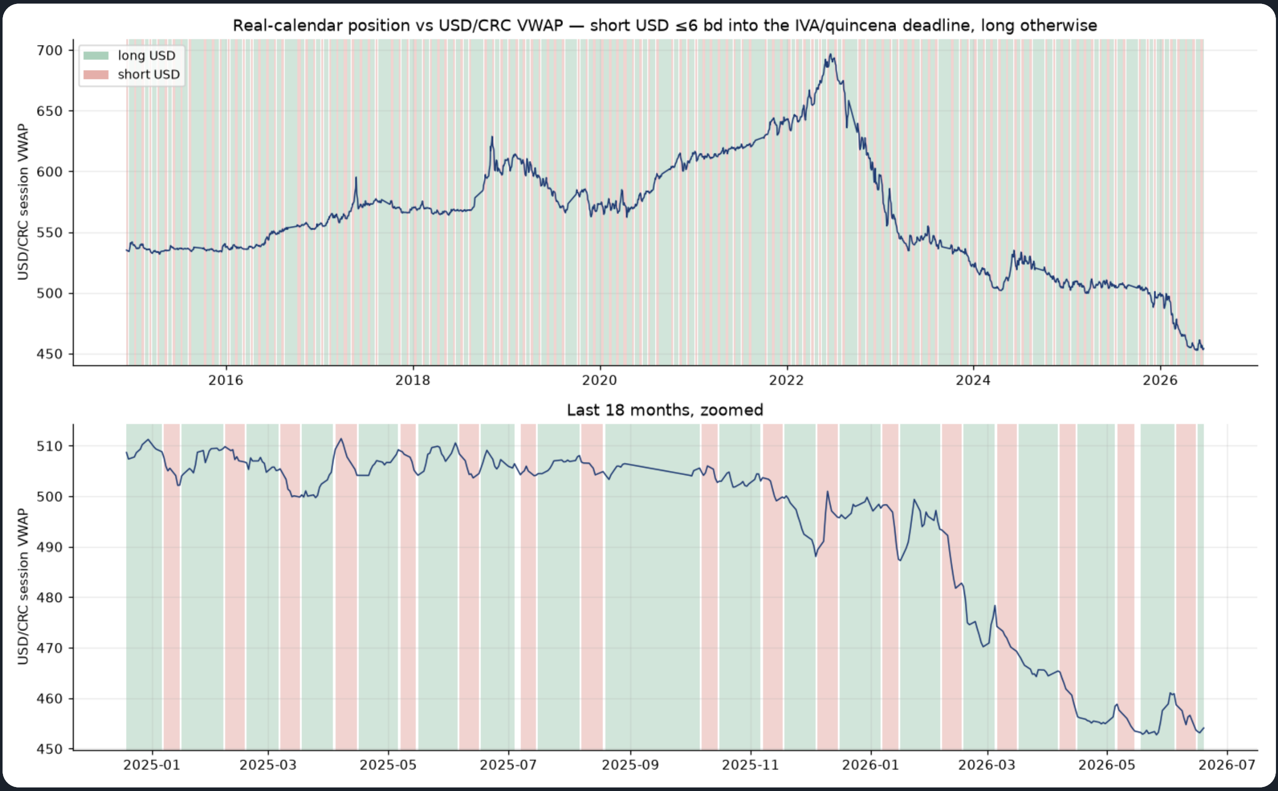


Red line = the in-sample / out-of-sample boundary; the bands are tuned only to the left of it. The overlay (green) tracks the baseline's compounding while its drawdowns (lower panel) are visibly shallower throughout.

But read the overlay as a RISK improvement, not more money. A paired t-test on the daily out-of-sample P&L difference gives t = -0.35, **p = 0.7292**: total P&L is statistically unchanged (indeed nominally $-27k lower out of sample). What the overlay buys is **+1.13 out-of-sample Sharpe and a ~45% cut in max drawdown** — a smoother path you can size up, not the same notional earning more. Volatility-scaled bands (trail 2.5× / floor 3.0× realised vol) scored *worse* than fixed bps in both windows and are reported as a negative result, not dropped.

A6. The position against the price

USD/CRC session VWAP shaded by the recommended calendar position



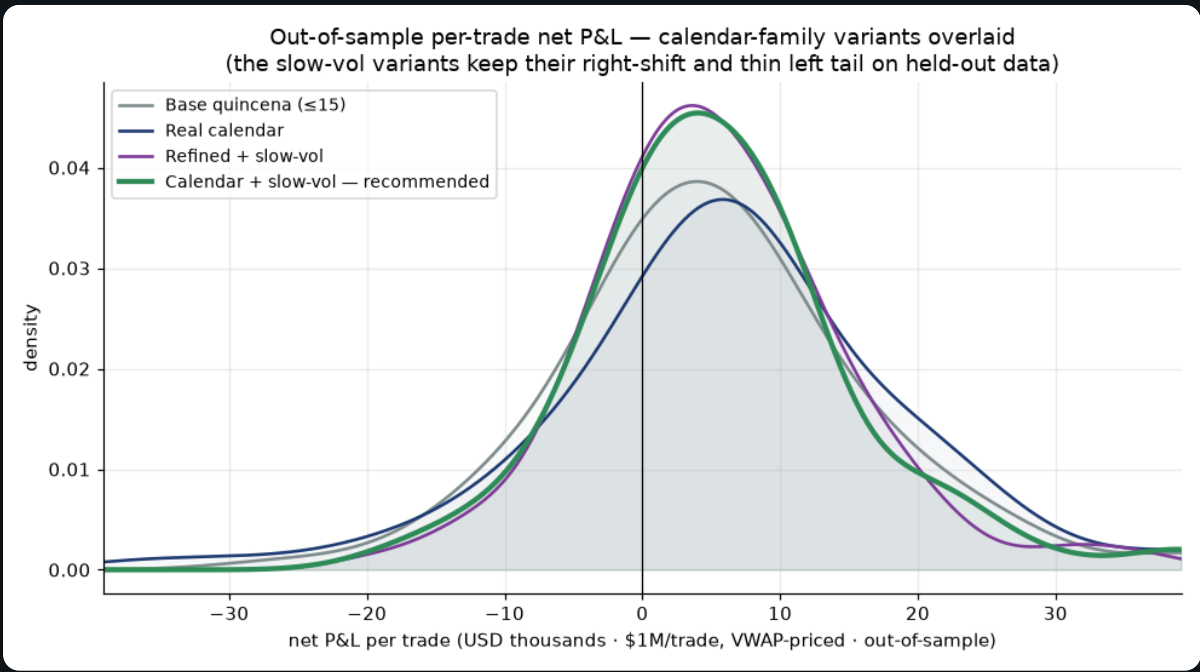
Green = long USD, red = short USD (short ≤6 business days into the IVA/quincena deadline). The entry is always in a position — there is no flat/no-trade state — so every session carries one shade or the other; the recommended rule additionally holds these at half size when the slow volume regime disagrees (same colour, thinner book). The short (red) bands cluster mid-month, exactly where the cash-flow supply surge lands.

A7. Return distributions — out-of-sample, per trade

Net P&L **per trade** (per directional roundtrip), in USD at $1M/trade, **VWAP-to-VWAP** — the P&L a desk actually books at each exit, not a day-by-day mark — and built on the **out-of-sample window only** (2021-11-18 → 2026-06-19, 103 held-out trades per variant). So these are the outcomes an operator would have booked on data the rule was never tuned on. Selection is on in-sample Sharpe; each panel reports IS→OOS Sharpe so you can see which edges survive. The recommended **Calendar + slow-vol** carries the badge: it is right-shifted, has the thinnest left tail of the family (out-of-sample skew **+0.74**, worst trade $-18k vs $-50k for the base rule), and it posts the family's best out-of-sample Sharpe, strengthening from 3.22 to 3.55 where the flat-sized entries decay.

Overlay (all four)

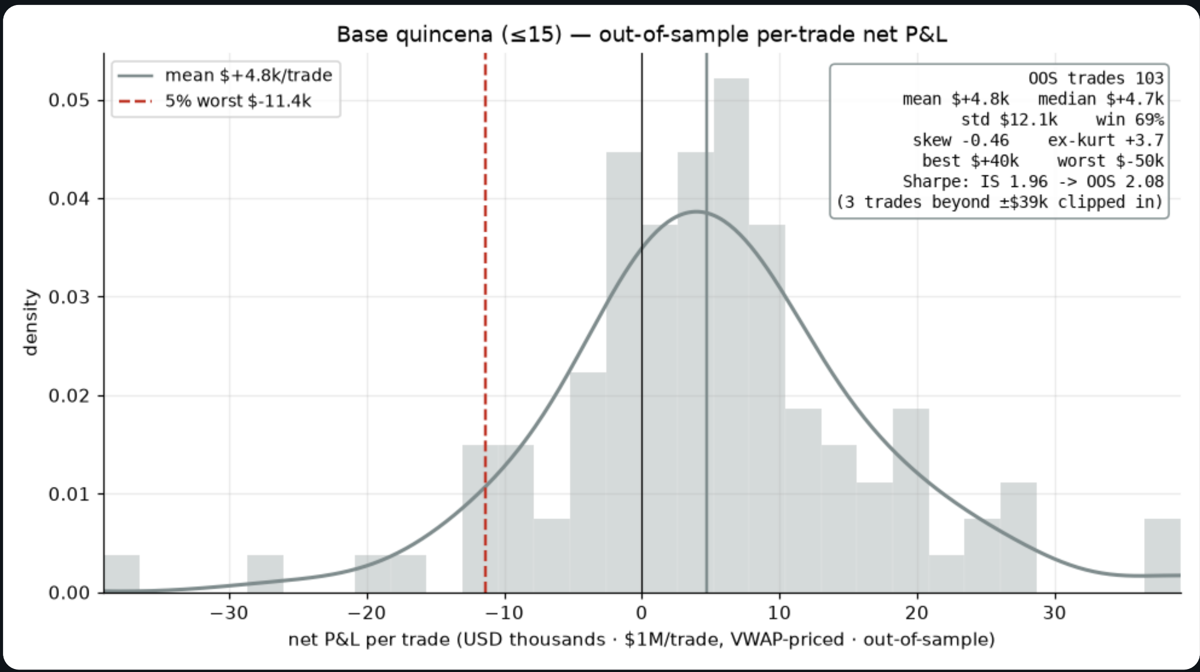
All four calendar-family variants, OOS per-trade P&L overlaid



On held-out data the two slow-vol variants stay peaked and right-shifted with a thin left tail; the flat-sized calendar and the crude base rule carry a heavier left tail. The recommended calendar + slow-vol curve is the rightmost mode.

Base quincena

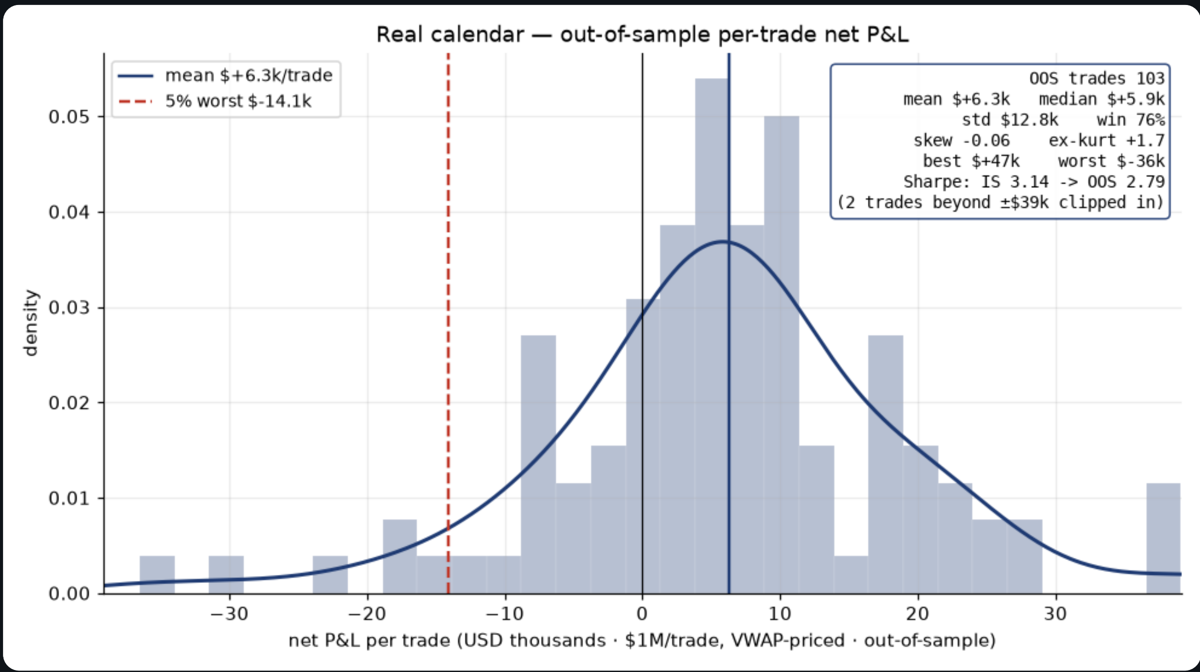
Base quincena (≤15) — out-of-sample per-trade net P&L



The crude ancestor: the widest spread of outcomes, a LEFT-skewed tail, and the heaviest OOS losers — the weakest risk-adjusted variant of the family throughout. 103 OOS trades · mean $+4.8k/trade, median $+4.7k, win 69% · std $12.1k · skew -0.46 · worst $-50k · Sharpe IS 1.96 → OOS 2.08.

Real calendar

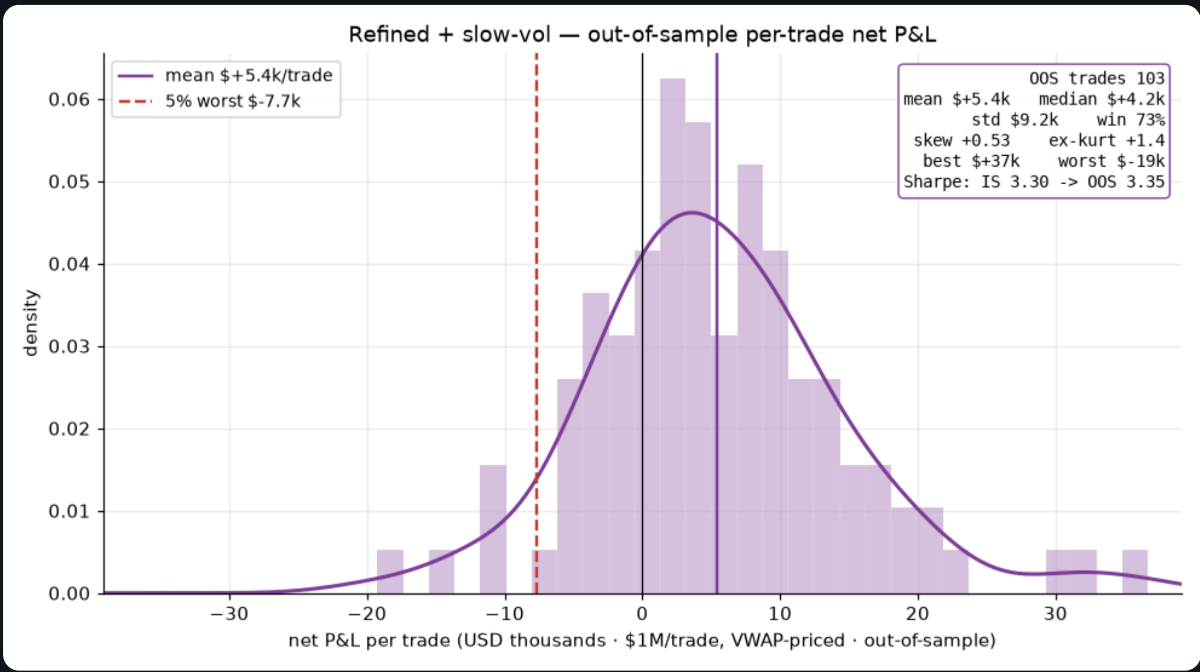
Real calendar (flat-sized) — out-of-sample per-trade net P&L



The deadline-anchored entry at flat size. Strong, but its left tail is intact and its OOS Sharpe slips below in-sample. 103 OOS trades · mean $+6.3k/trade, median $+5.9k, win 76% · std $12.8k · skew -0.06 · worst $-36k · Sharpe IS 3.14 → OOS 2.79.

Refined + slow-vol

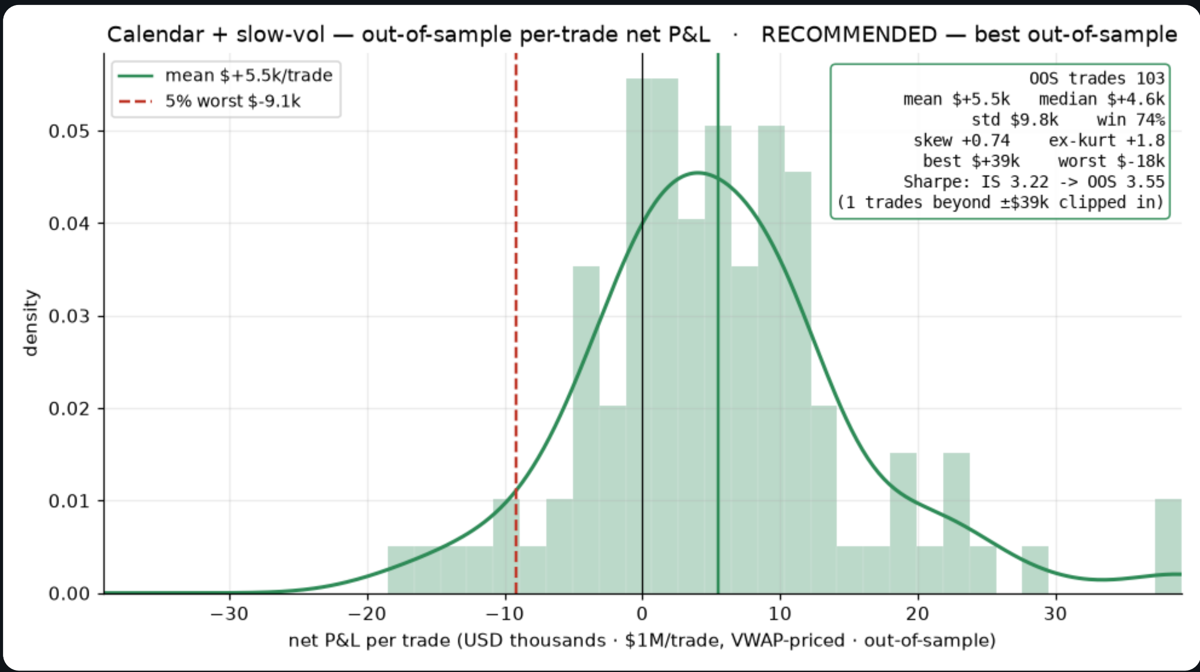
Refined (fixed 5–15) + slow-vol — out-of-sample per-trade net P&L



The slow-volume size trim on the FIXED 5–15 window — the trim cuts the fat losing trades, flipping the skew positive and holding the OOS Sharpe. 103 OOS trades · mean $+5.4k/trade, median $+4.2k, win 73% · std $9.2k · skew +0.53 · worst $-19k · Sharpe IS 3.30 → OOS 3.35.

Calendar + slow-vol

Calendar + slow-vol — out-of-sample per-trade net P&L



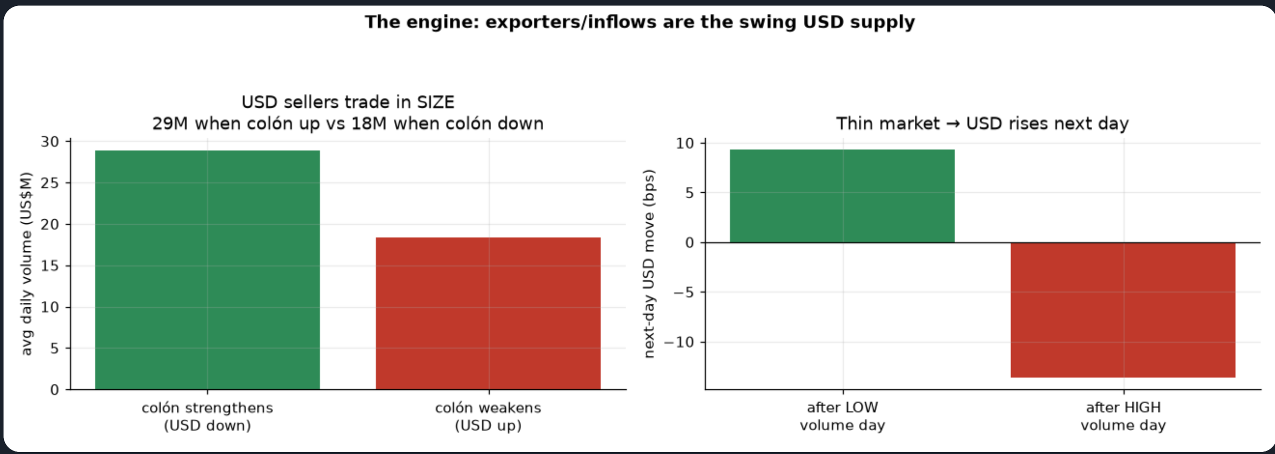
THE RECOMMENDED RULE: deadline-anchored entry + slow-volume trim. Right-shifted, the thinnest left tail of the family, positive skew, and the family's best out-of-sample Sharpe — it strengthens out of sample rather than giving the edge back. 103 OOS trades · mean $+5.5k/trade, median $+4.6k, win 74% · std $9.8k · skew +0.74 · worst $-18k · Sharpe IS 3.22 → OOS 3.55.

VARIANT · NET P&L PER TRADE (OUT-OF-SAMPLE), $1M/TRADE, VWAP-PRICED	OOS TRADES	MEAN/TRADE	MEDIAN	STD	WIN %	SKEW	WORST TRADE	IS → OOS SHARPE
Base quincena (≤15)	103	$+4.8k	$+4.7k	$12.1k	69%	-0.46	$-50k	1.96 → 2.08
Real calendar	103	$+6.3k	$+5.9k	$12.8k	76%	-0.06	$-36k	3.14 → 2.79
Refined + slow-vol	103	$+5.4k	$+4.2k	$9.2k	73%	+0.53	$-19k	3.30 → 3.35
Calendar + slow-vol RECOMMENDED	103	$+5.5k	$+4.6k	$9.8k	74%	+0.74	$-18k	3.22 → 3.55

PART B · THE UNDERLYING DYNAMICS — WHAT WE ARE EXPLOITING

B1. The engine: exporters are the swing USD supply

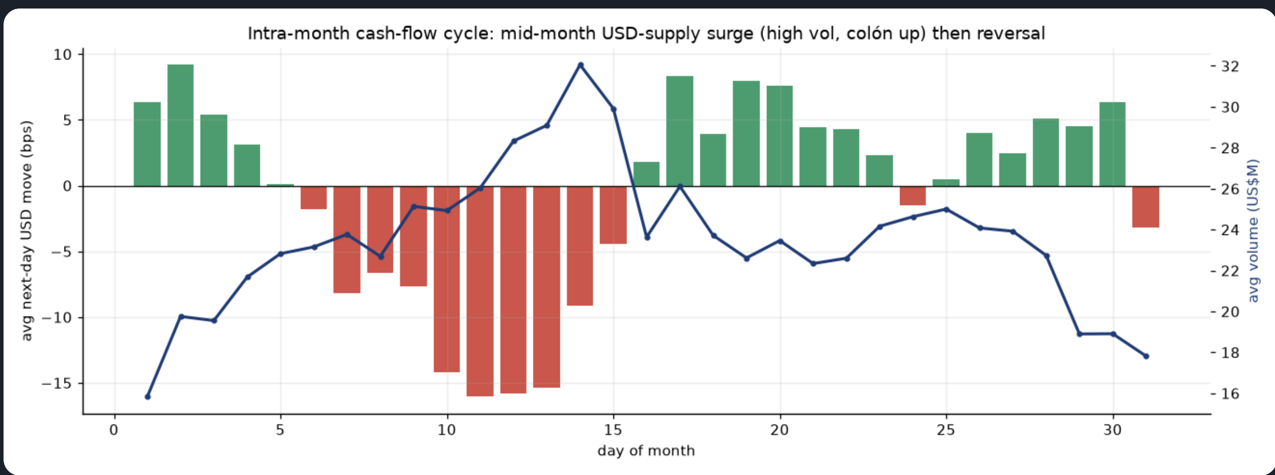
Volume when the colón strengthens vs weakens, and next-day move by volume



On days the colón strengthens (USD falls) the market trades 29M — far more than the 18M on days it weakens. USD sellers (exporters, remittances, FDI) come in size; USD buyers (importers) trickle. So HIGH volume = supply present = colón up, and LOW volume = thin market = USD drifts up next day (+9.3 bps after quiet days vs -13.7 after busy ones). That asymmetry IS the volume signal.

B2. The intra-month cash-flow cycle (why the calendar works)

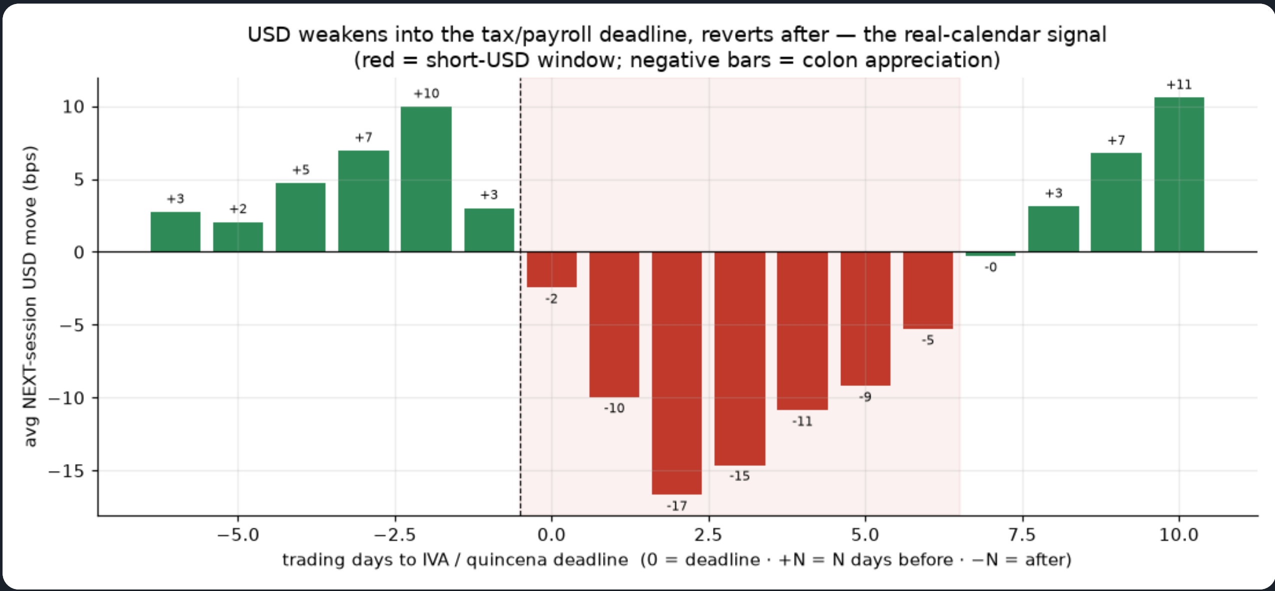
Average next-day USD move (bars) and volume (line) by day-of-month



A clear monthly rhythm: volume peaks mid-month (~day 13–15) and the colón strengthens hardest right then (red bars) — a recurring USD-supply surge (exporter settlements, tax/paydate conversions). The first days and the second half of the month lean the other way (USD up). This is the quincena effect, and it is a cash-flow cycle, not a chart pattern.

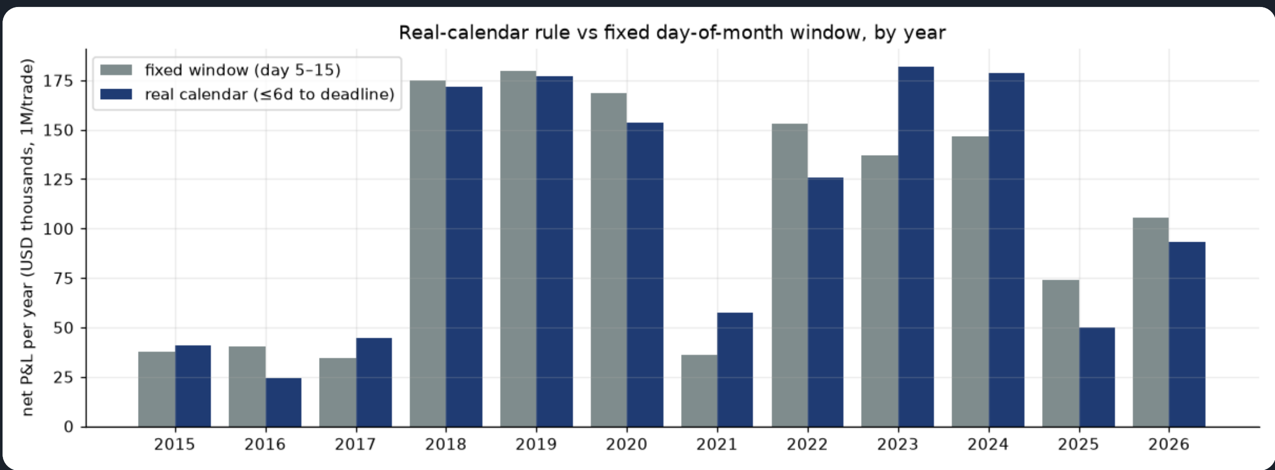
B2b. Aligning to the real deadline sharpens the signal

Average next-day USD move by business-days-to the IVA/quincena deadline



Re-indexing the same data from raw day-of-month to *trading days to the statutory IVA (15th) / mid-month payroll deadline* makes the mechanism unmistakable: the colón strengthens hardest 1–5 business days BEFORE the deadline (−12 to −16 bps next-day), then reverts to USD-up right after — a textbook conversion-then-clearing pattern. The red band is the short-USD window. Because the anchor rolls with weekends and holidays, the calendar rule (Sharpe 2.88, a broad plateau of 1.45–2.88 across lookbacks) edges the fixed-day version while explaining the flow.

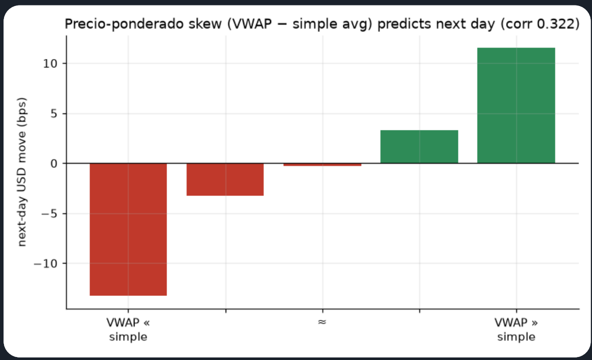
Net P&L by year — fixed day-of-month window vs real calendar ($1M/trade)



Deadline-anchoring holds up year by year (worst year $24k), not just in aggregate.

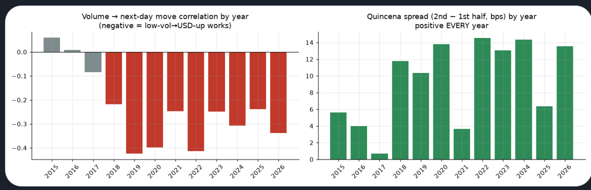
B3. Two more relationships

Precio-ponderado skew (VWAP – simple average)



When large tickets pull the weighted price above the simple average, USD tends to keep rising next day (corr 0.322). It reads large-player direction — but, as the dollar table shows, it flips daily and is too costly to trade alone.

Year-by-year stability of both effects



The volume → move link is negative every year since 2018; the quincena spread is positive in ALL 12 years. That cross-regime persistence is why we treat these as structural, not curve-fit.

PART C · HOW EACH RULE DECIDES (EXACT LOGIC)

The recommended calendar rule is the default tab; the secondary signals are one click away.

Calendar rule

Calendar + slow-vol rule — the recommendation

```
≤ 6 business days before the IVA/quincena deadline (through it) → SHORT USD $1M
   (firms sell USD to raise colones for the 15th tax + payroll → colón strengthens)
otherwise → LONG USD $1M (supply fades, USD drifts up)
size trim: hold HALF size whenever a slow 20-day volume regime disagrees with the trade
   (full size only when supply/thinness confirms the direction)
```

Why: the intra-month cash-flow cycle (Part B) — companies convert USD→CRC for the D-104 IVA filing (due the 15th, rolled to the next business day) and the mid-month quincena payroll, a recurring supply surge that strengthens the colón into the deadline and reverts after. The *real-calendar* anchor rolls with weekends/holidays (economically the right entry); the *slow-vol trim* only cuts SIZE (never the direction) when the volume regime disagrees, which removes the fat losing trades. **Result (VWAP-priced):** $95k/yr, full-sample Sharpe 3.31, ~18 trades/yr, max drawdown $-37k — and, crucially, an **out-of-sample Sharpe of 3.55** ($123k/yr on the 40% held out) — the family's best, and it strengthens out of sample where the strong flat-sized entries give the edge back (A4/A7). The base "short the whole first half" rule ($80k/yr, Sharpe 1.97) is the crude ancestor this refines.

Volume

Volume — "buy USD when the market is quiet"

```
seasonal_vol = today_volume ÷ (avg volume on the same weekday, to date)
if seasonal_vol < 1.0 → LONG USD $1M (quieter than normal)
if seasonal_vol ≥ 1.0 → SHORT USD $1M (busier than normal)
```

Why: quiet = exporters absent = importer demand lifts USD. Dividing by the same-weekday average makes "quiet" mean quiet *for a Tuesday*. **Result:** directionally right but flips ~71.2×/yr — even VWAP-priced it only nets Sharpe 1.06 standalone. Best used as the slow regime filter on the calendar trade.

Precio-ponderado skew

Precio-ponderado skew — "follow the big tickets"

```
if VWAP > simple_average → LONG USD $1M (large trades lifted the price)
if VWAP ≤ simple_average → SHORT USD $1M
```

Why: the weighted-vs-simple gap reveals whether large players were buying or selling USD (corr 0.322 with the next-day move). **Result:** flips ~93.7×/yr → Sharpe 0.93 net; a tie-breaker, not a standalone book.

Combined & ML

Combined vote & the ML model

```
Combined = majority vote of the three signals (always ±1) → $1M long/short
ML = logistic P(USD up) on all features; position = clip((P − 0.5) × 5, −1, +1)
```

Why the ML sizes by conviction: a raw daily flip trades ~95×/yr and dies on cost. Scaling the position by confidence keeps turnover near 54.7×/yr while still capturing the edge. On the realistic VWAP basis the combined vote nets Sharpe 2.03 and the ML 3.84 — but both lean on the same calendar + volume economics as the simple rule above.

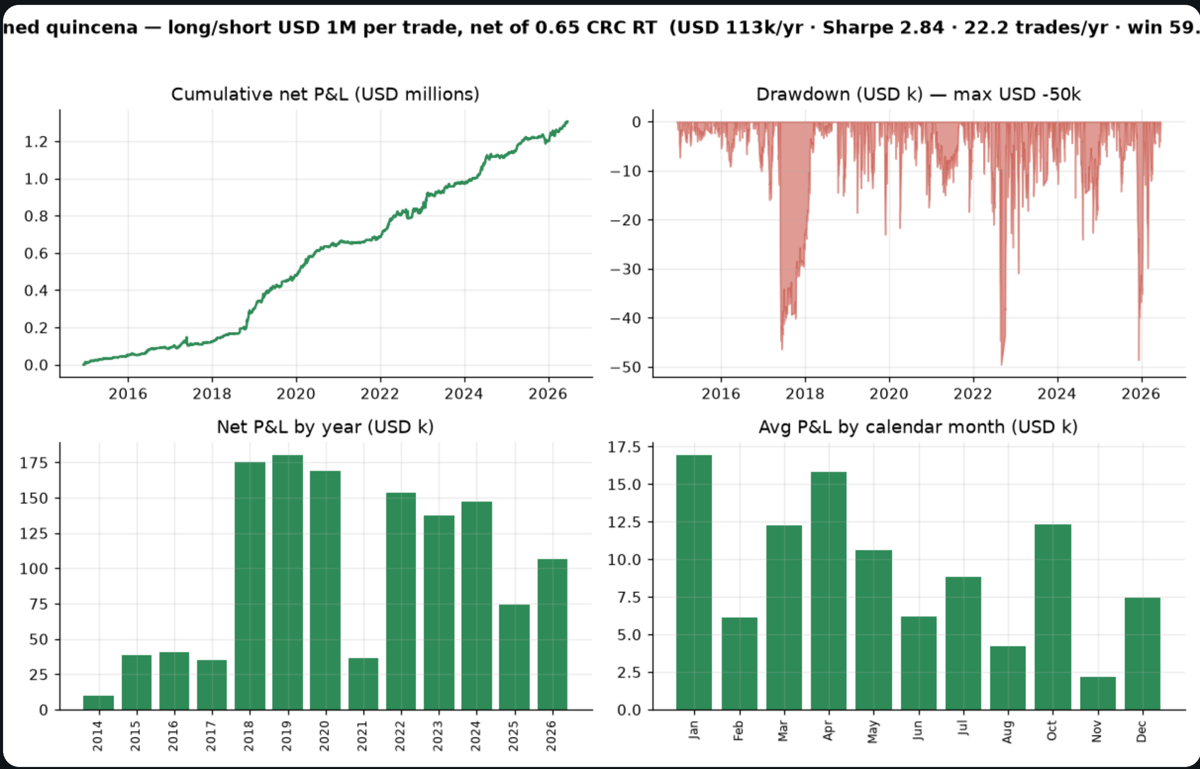
PART D · RESULTS AT $1,000,000 PER TRADE (VWAP-PRICED, NET OF COST)

The recommended calendar family sits at the top; the raw building-block signals and the ML model follow. Equity curves are tabbed — recommended first.

STRATEGY · $1M/TRADE · VWAP-PRICED, NET 0.65 CRC RT	PER YEAR	SHARPE (FULL)	SHARPE (OOS)	TRADES/YR	WIN %	MAX DD
RECOMMENDED — CALENDAR / QUINCENA FAMILY (OOS = LAST 40%, HELD OUT)						
Calendar + slow-vol RECOMMENDED · BEST OOS	$95k	3.31	3.55	18.2	59.5	$-37k
Real calendar (≤6 bd), flat size	$114k	2.88	2.79	22.2	59.8	$-48k
Refined (fixed 5–15) + slow-vol	$94k	3.23	3.35	18.2	59.2	$-37k
Base quincena (short ≤15)	$80k	1.97	2.08	22.2	56.2	$-62k
BUILDING-BLOCK SIGNALS — TRADE GENTLY / AS FILTERS						
Volume rule	$43k	1.06	—	71.2	50.5	$-242k
Precio-ponderado skew rule	$38k	0.93	—	93.7	49.5	$-397k
Combined vote (3 signals)	$82k	2.03	—	78.1	53.6	$-278k
ML combined model (conviction-sized, walk-forward)	$122k	3.84	—	54.7	52.3	$-36k

Recommended calendar

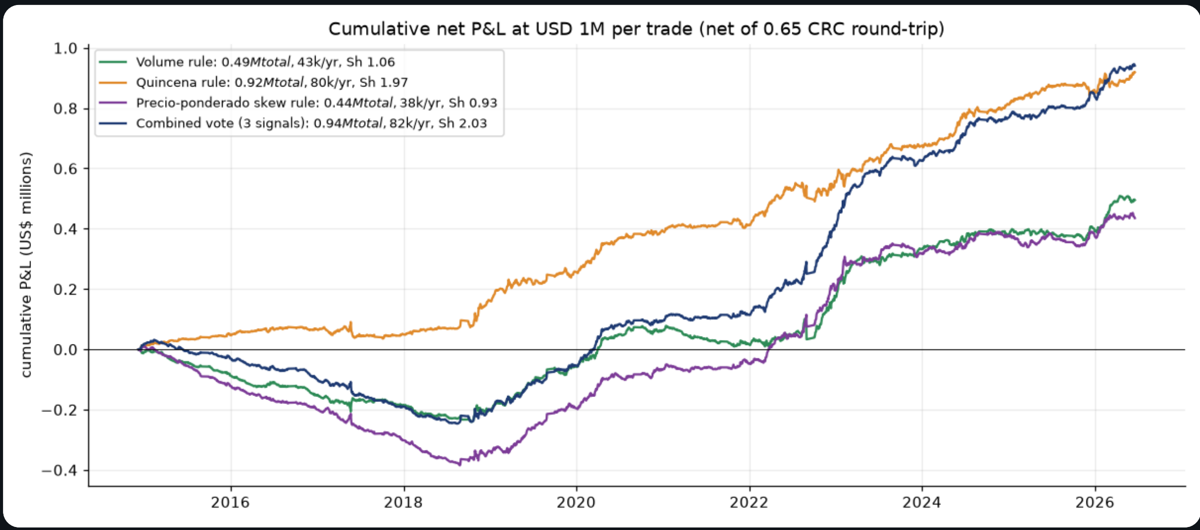
Recommended calendar rule — equity, drawdown, yearly & monthly P&L



Smooth compounding at ~$114k/yr with shallow drawdowns; every year and (on average) every month positive — January and April strongest. VWAP-priced, net of cost.

Building-block rules

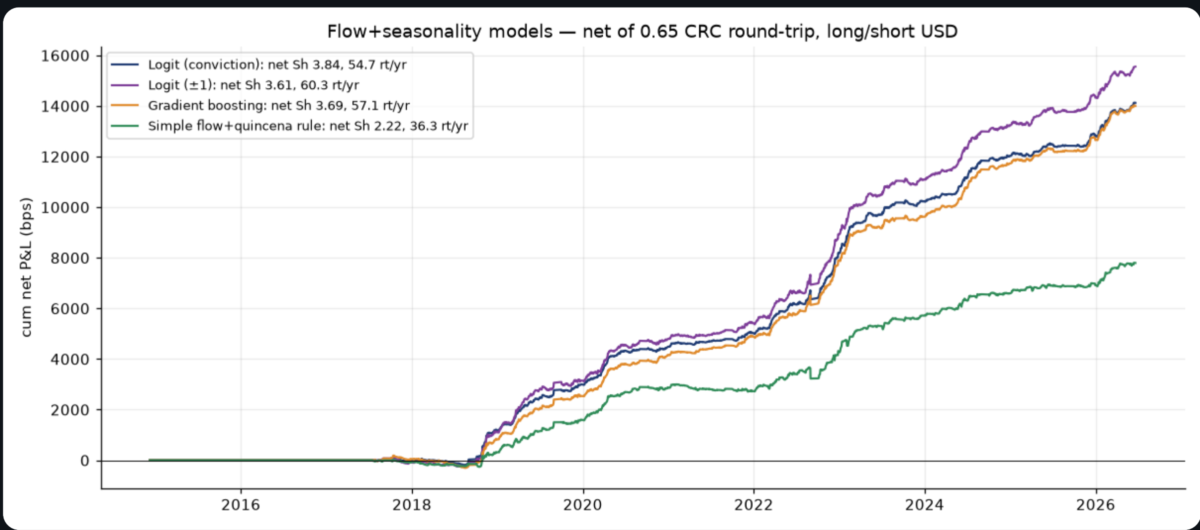
Cumulative net P&L in US$ millions — building-block rules, $1M per trade



The base quincena (green) compounds smoothly; the combined vote is decent but dragged by the turnover of the volume and skew legs. The volume rule is directionally right but churns — the signal is real, the daily flipping is the cost problem.

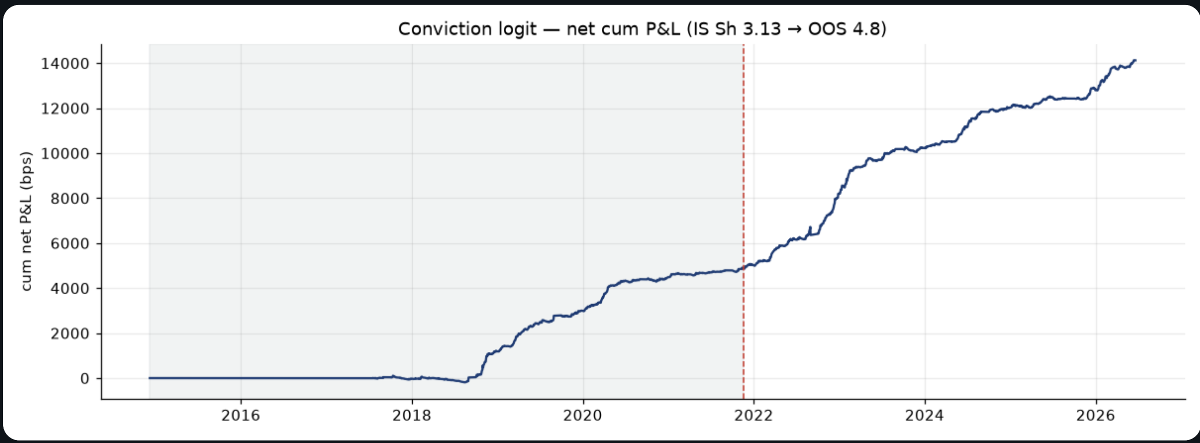
ML model

Conviction-sized ML model — cumulative net P&L (walk-forward, VWAP-priced)



Every prediction is out-of-sample (walk-forward). Conviction sizing keeps turnover near 54.7×/yr; net Sharpe 3.84. It mostly repackages the same calendar + volume economics as the simple rule, at higher complexity.

ML: in-sample vs out-of-sample (net)



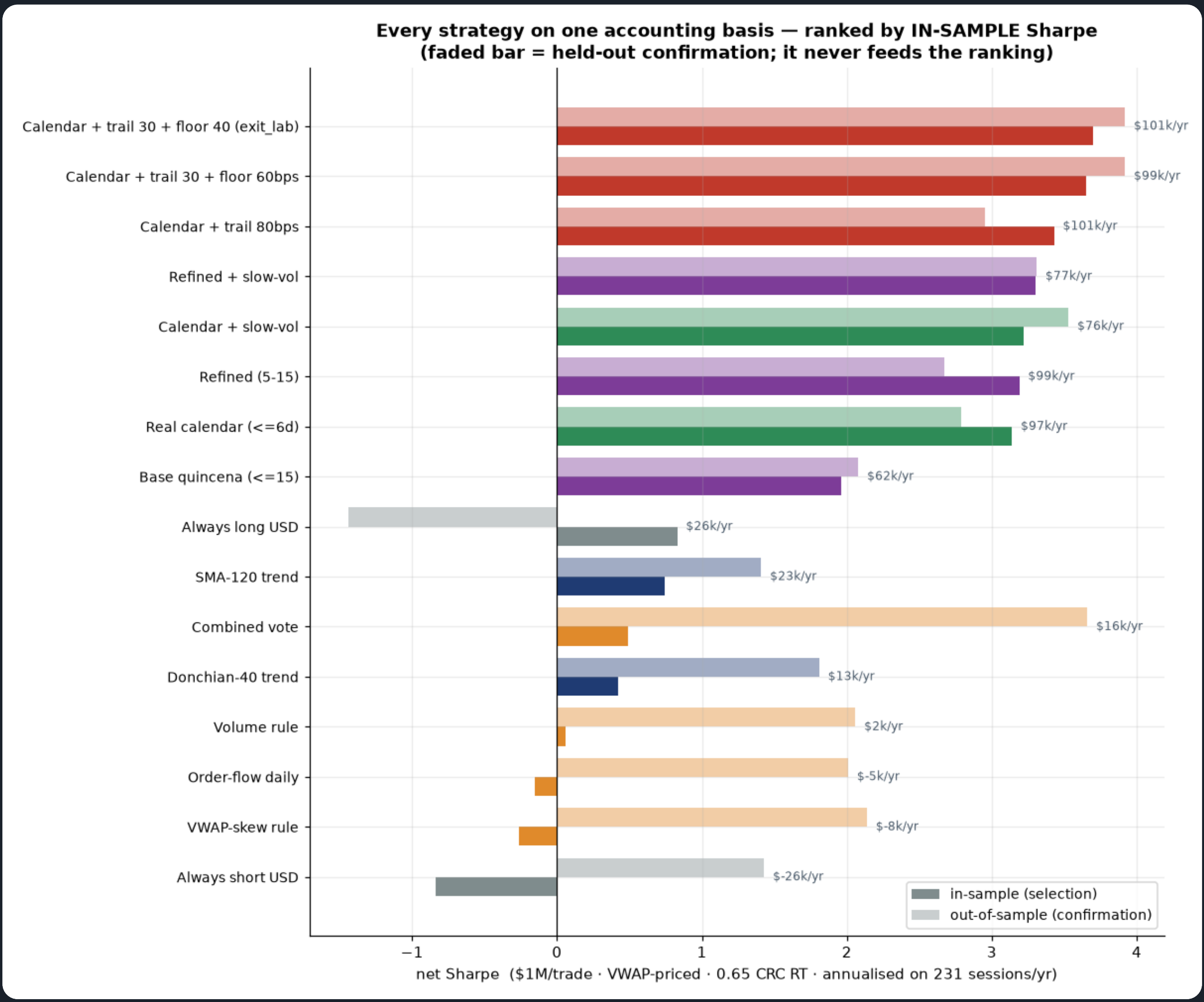
OOS Sharpe 4.8 vs IS 3.13; a 5-day feature lag collapses accuracy (no look-ahead), permutation p=0.0.

PART R · UNIFIED RANKING — EVERY STRATEGY ON ONE HONEST BASIS (NEWEST RUN)

> Until this run each module carried its own accounting, so the headline numbers were **not comparable** — `strategies.py` reported gross Sharpe, `backtest.py` priced close-to-close in bps, the calendar modules priced VWAP in dollars. `src/rank_strategies.py` re-implements **every** rule against ONE frame and ONE basis — VWAP-to-VWAP, $1M/trade, 0.325 CRC per side, annualised on the real **231 sessions/yr** — and ranks by **in-sample** Sharpe (first 60% of history, split 2021-11-18). The out-of-sample column is carried for honesty and never feeds the ordering. This table **supersedes the per-module tables** for cross-strategy comparison.

#	STRATEGY · VWAP-PRICED, NET 0.65 CRC RT, 231 SESS/YR	FAMILY	IS SHARPE	OOS SHARPE	IS $/YR	IS MAX DD
1	Calendar + trail 30 + floor 40 (exit_lab) TOP OF BOOK	exit	3.7	3.92	$101k	$-16k
2	Calendar + trail 30 + floor 60bps	exit	3.65	3.92	$99k	$-18k
3	Calendar + trail 80bps	exit	3.43	2.95	$101k	$-23k
4	Refined + slow-vol	calendar	3.3	3.31	$77k	$-25k
5	Calendar + slow-vol	calendar	3.22	3.53	$76k	$-24k
6	Refined (5-15)	calendar	3.19	2.67	$99k	$-46k
7	Real calendar (<=6d)	calendar	3.14	2.79	$97k	$-45k
8	Base quincena (<=15)	calendar	1.96	2.08	$62k	$-55k
9	Always long USD	benchmark	0.83	-1.44	$26k	$-111k
10	SMA-120 trend	trend	0.74	1.41	$23k	$-66k
11	Combined vote	flow	0.49	3.66	$16k	$-278k
12	Donchian-40 trend	trend	0.42	1.81	$13k	$-128k
13	Volume rule	flow	0.06	2.06	$2k	$-242k
14	Order-flow daily	flow	-0.15	2.01	$-5k	$-273k
15	VWAP-skew rule	flow	-0.26	2.14	$-8k	$-397k
16	Always short USD	benchmark	-0.84	1.43	$-26k	$-195k

Every strategy ranked on one common accounting basis (in-sample net Sharpe)



The calendar family (green) owns the top of the book and the flow family (red) owns the bottom — in-sample. The top three seats are the calendar entry plus an exit overlay; the recommended trail 30 + floor 40 leads at IS Sharpe 3.7 → OOS 3.92.

The calendar family owns the top, the flow family owns the bottom — in-sample. Every flow rule (order-flow, volume, skew, combined vote) is worthless-to-negative on 2014–2021 and only works on 2021–2026. Their full-sample numbers average a dead regime with a live one, which is exactly why they read as mediocre-but-real rather than regime-contingent.

The flow rules carry catastrophic drawdowns — $-397k to $-278k in-sample against the calendar rule's $-16k–$-55k, at 80–97 round trips/yr. Even where the out-of-sample Sharpe looks excellent (Combined vote 3.66), the path is not something a desk would fund.

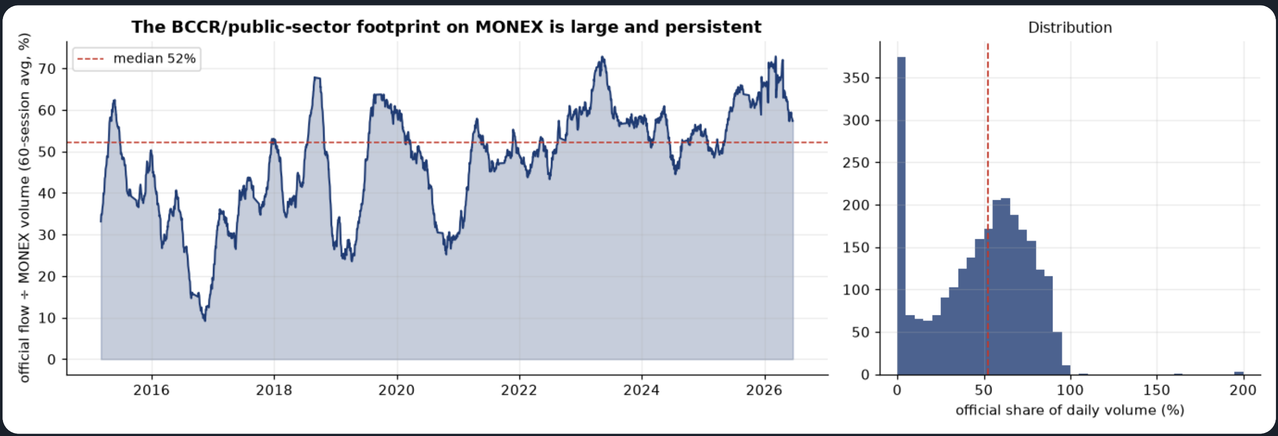
Sanity check on the harness: "Always long USD" (IS 0.83 / OOS -1.44) and "Always short USD" (IS -0.84 / OOS 1.43) are near-exact mirrors, as two opposite always-on positions must be — confirming the P&L and cost accounting.

Honest caveat on the metric. Because the in-sample window (2014–2021) is the pegged/low-vol regime and the out-of-sample window (2021–2026) is the float, in-sample selection structurally penalises any regime-contingent signal. The calendar rule wins partly *because* it is the one edge that works in both — a real virtue, but not the same claim as "the flow signals don't work."

PART I · THE BCCR'S HAND — OFFICIAL FLOW & RESERVES (NEWEST DATA)

MONEX volume is **unsigned** — it says how much traded, not who was buying. Two BCCR exports (daily FX **intervention**, CodCuadro 1587; month-end **net reserves**, CodCuadro 8) fill that gap. The official side — the BCCR plus the non-bank public sector's (RECOPE et al.) USD requirement routed through MONEX — is **a median 52% of daily MONEX volume** (89% of sessions carry official flow). So the **differential** (total MONEX minus official) is what the private market actually did, and the signed official flow is information the volume series cannot carry. This section adds that data and asks whether it beats the published rule.

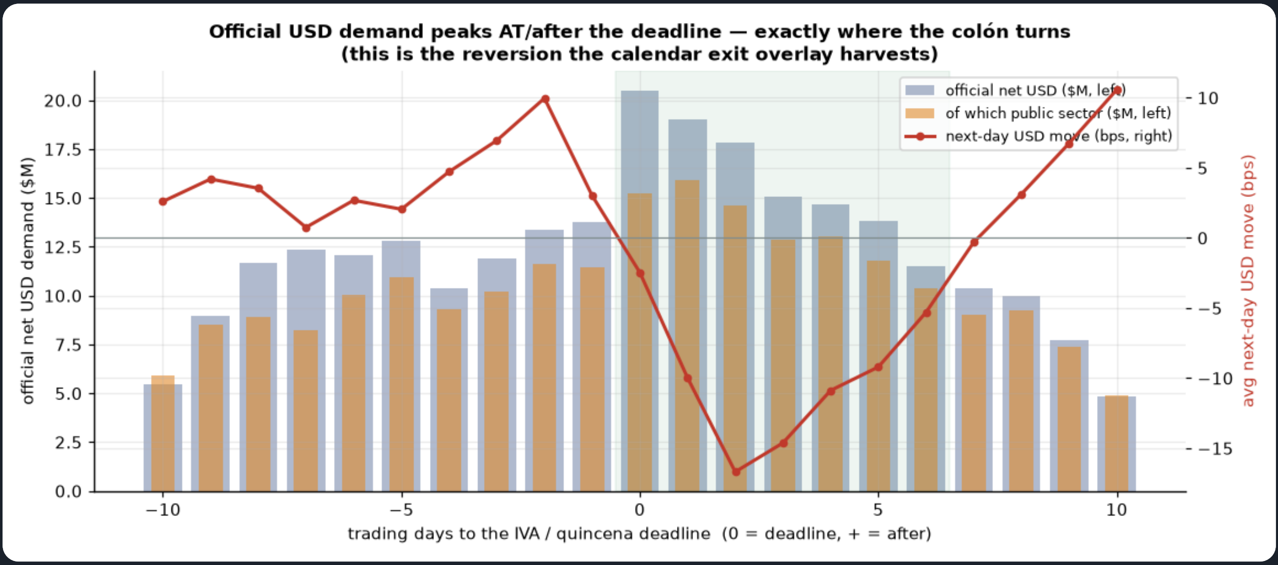
Official (BCCR + public-sector) flow as a share of MONEX volume



The official footprint is large and persistent — a 60-session average around 48% of traded volume, so on a typical day roughly half of what crosses MONEX is the BCCR or the public sector, not private supply/demand. Any read of 'private' flow from raw volume is therefore contaminated by an official component of this size.

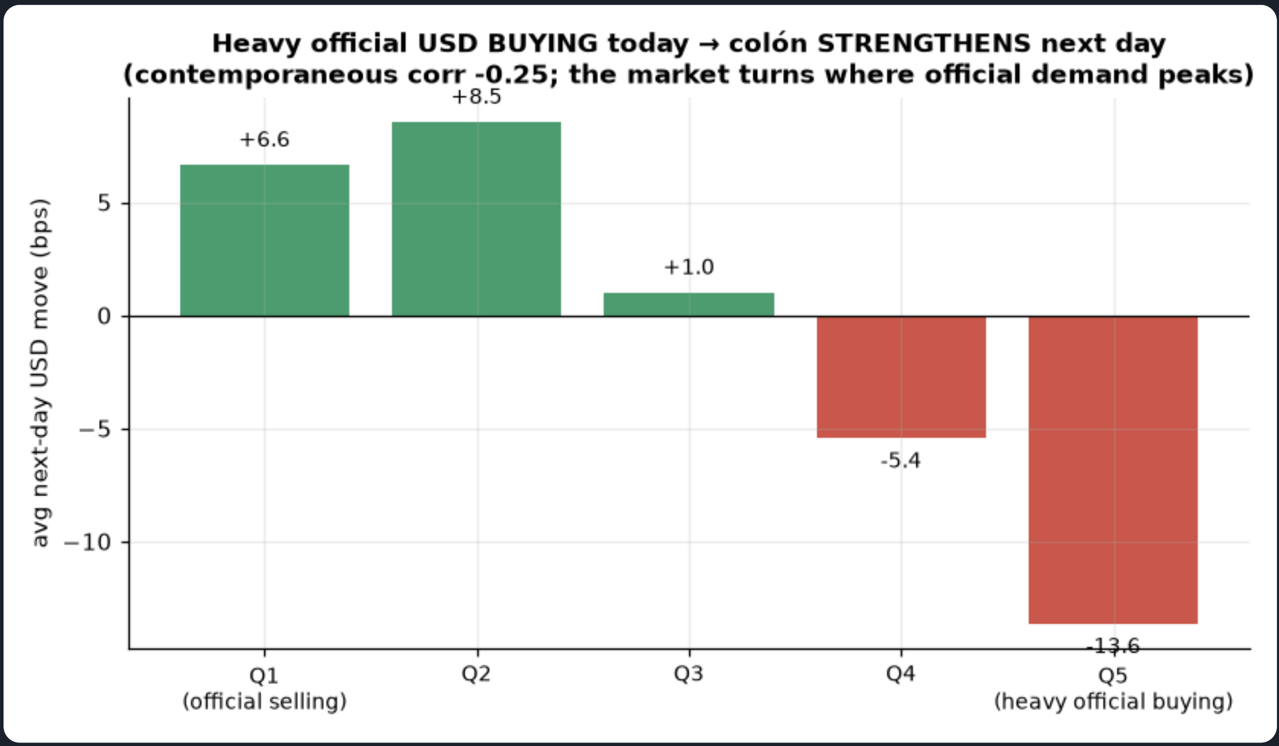
I1. The mechanism — official demand marks where the colón turns

Official USD demand vs the next-day move, by trading-days-to the deadline



Official (and specifically public-sector) USD demand PEAKS at and just after the IVA/quincena deadline — the shaded window — which is exactly where the colón strengthens hardest next-day (the red line dives to -14-bps-class moves). Heavy official demand does not push USD up; it marks the point where private supply overwhelms and the market reverts. This is the same post-deadline reversion the exit overlay (A5b) harvests — now with a name.

Next-day USD move by official net-buying quintile



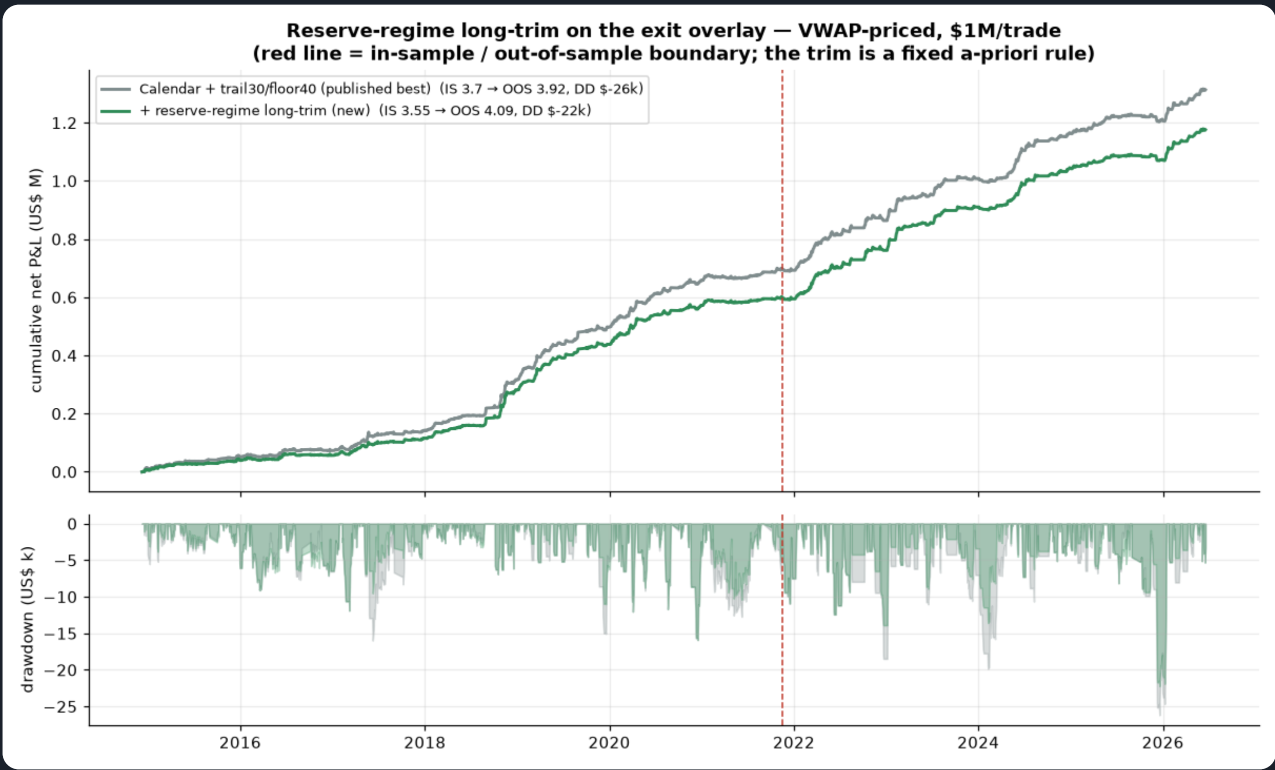
The signed signal volume cannot see: the heaviest official-net-buying days are followed by the largest USD-DOWN moves (contemporaneous corr -0.25). It is a real reversion signal, but it decays sharply when lagged one session (-0.08) and depends on same-day disclosure of the intervention figures — so it is reported as mechanism, not used as a live entry signal.

I2. Does it beat the published rule? A reserve-regime size trim

> The tradeable win is not the daily flow (redundant with volume, and disclosure-sensitive) but the **reserves**. When BCCR reserves are **rising** — the bank accumulating USD, a colón-supportive regime — we trim long-USD size to half. It is strictly causal (a lagged monthly figure, no disclosure issue) and a fixed a-priori rule. Stacked on the published exit overlay it lifts out-of-sample Sharpe **3.92 → 4.09** and cuts max drawdown **$-26k → $-22k**. Reserves rose in only 50% of months (not a one-way trend), so this is a regime read, not a disguised "hold less long USD".

CALENDAR RULE · $1M/TRADE · VWAP-PRICED, NET COST	IS SHARPE	OOS SHARPE	FULL $/YR	MAX DD	NOTE
Calendar + trail30/floor40	3.7	3.92	$114k	$-26k	published best (A5b)
+ reserve-regime long-trim NEW · BEST OOS/DD	3.55	4.09	$102k	$-22k	trim longs when reserves rising
+ official-flow trim (lagged)	3.37	4.07	$84k	$-21k	strictly-lagged official regime
CONTROL: blind long-trim	3.62	4.1	$86k	$-21k	trim ALL longs — no BCCR data

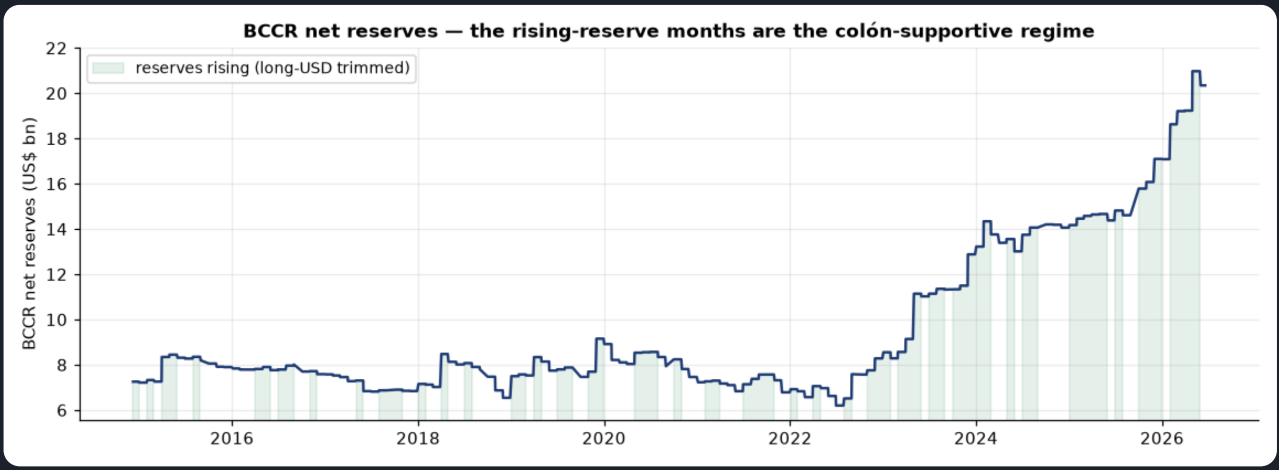
Exit-overlay baseline vs + reserve-regime long-trim — equity & drawdown



The trim (green) tracks the baseline's compounding while its drawdowns (lower panel) are visibly shallower, most clearly through the deep late-sample selloff. Red line = the in-sample / out-of-sample boundary; the rule is fixed a-priori, nothing is fit on the test window.

The reserve signal does real work — it is not just 'trim longs'. A blind control that halves *every* long reaches a similar out-of-sample Sharpe, but the reserve-selective trim earns **$56k more out-of-sample** at the same risk reduction (paired t = 3.31, **p = 0.001**) — it keeps full size on the longs reserves say are safe and cuts only the ones it flags. Like the exit overlay, this is a **risk** improvement (lower drawdown, higher Sharpe, slightly less total P&L), not an alpha one. The official-flow trim is an equally-causal alternative that lands in the same place; the two BCCR series confirm each other.

BCCR net reserves and the rising-reserve (colón-supportive) regime

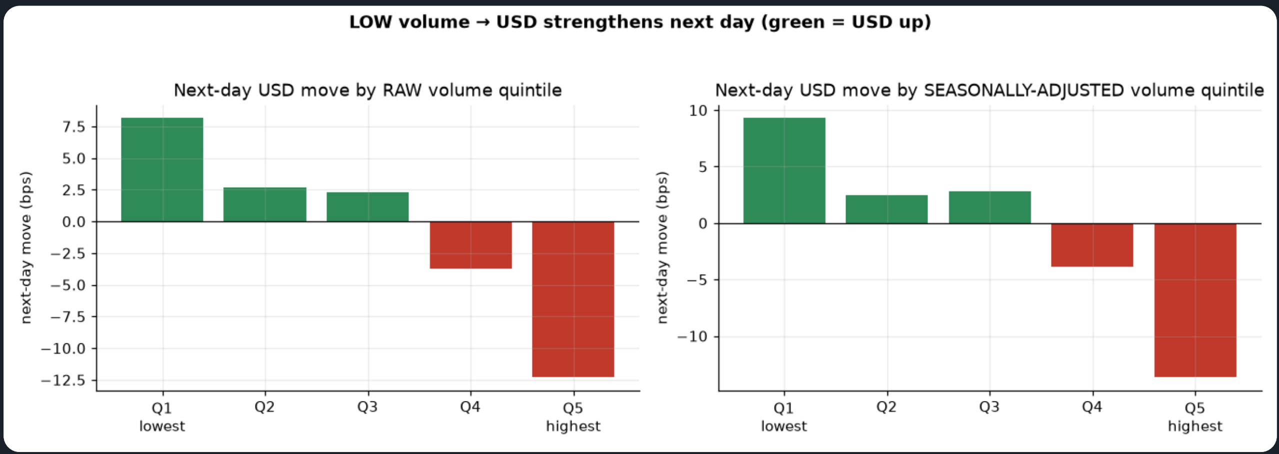


Reserves ran from $6.2bn to $21.0bn over the sample; the shaded rising-reserve months are when long-USD size is trimmed. The month-to-month change is roughly even (50% rising), which is why the trim is a genuine regime read.

PART E · RAW RELATIONSHIPS BEHIND THE SIGNALS

E1. Low volume → USD up (seasonally adjusted)

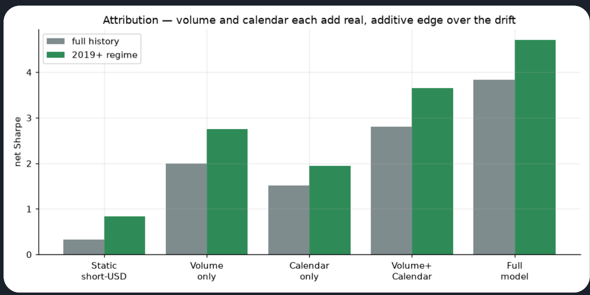
Next-day USD move by volume quintile



Monotonic: quietest days → USD up, busiest → USD down. Seasonal adjustment sharpens the extremes. This is the raw relationship the volume filter and the ML model lean on.

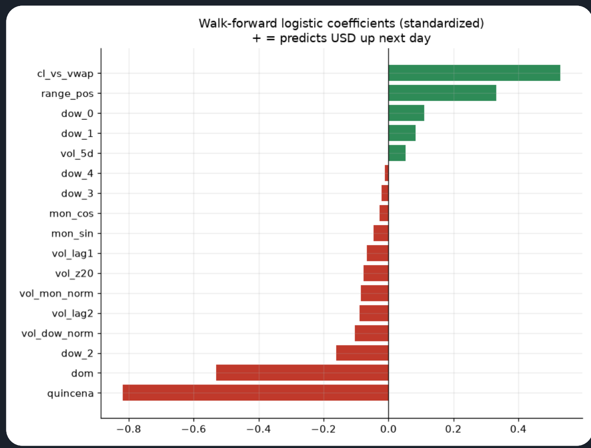
E2. Attribution & drivers

Net Sharpe by feature family



Volume-only and calendar-only each beat the drift and are additive — the calendar carries the most and volume refines it, which is exactly why the recommended rule pairs the calendar entry with a slow-volume size trim.

Walk-forward model coefficients



Quincena/day-of-month and volume dominate; no technical indicators used.

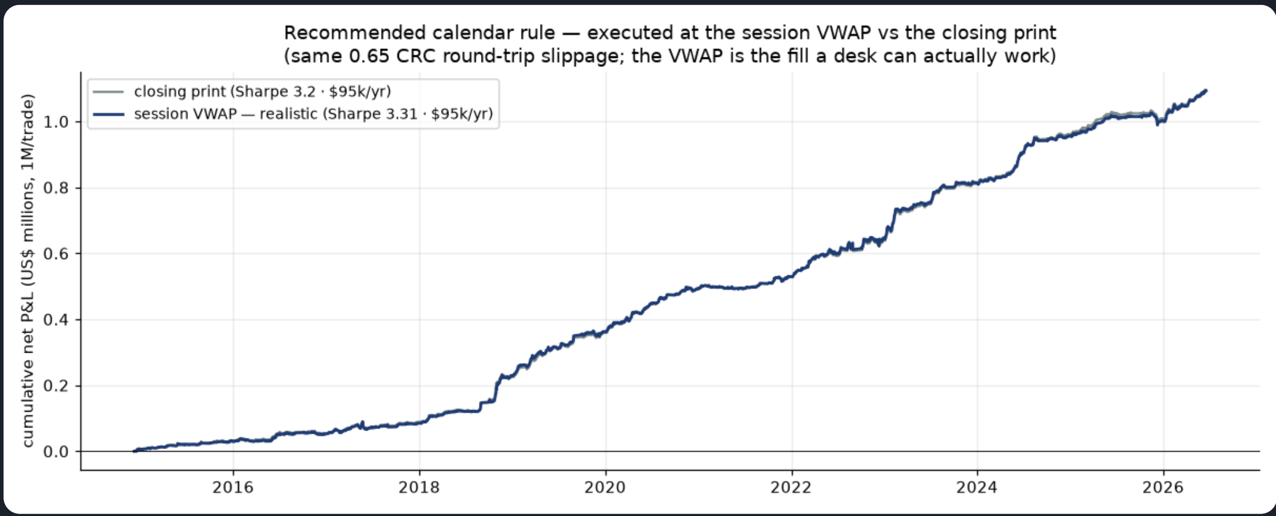
PART F · EXECUTION REALISM — VWAP VS THE CLOSING PRINT

F1. The recommended rule does not depend on catching the close

Every result in this report is already priced **VWAP-to-VWAP** — the fill a desk can actually work by spreading an order across the session, rather than the optimistic assumption of trading the last print. Here is the proof for the strategy we care about: the recommended calendar + slow-vol rule, same 0.65 CRC round-trip slippage, marked at the closing print vs the session VWAP. The VWAP fill **does not haircut the edge — it slightly improves the Sharpe** (3.2 → 3.31), because the VWAP is a steadier reference than a single closing tick.

RECOMMENDED CALENDAR RULE · $1M/TRADE · NET 0.65 CRC RT	PER YEAR	SHARPE	MAX DD
Closing-print fill (optimistic mark)	$95k	3.2	$-34k
Session VWAP fill USED EVERYWHERE IN THIS REPORT	$95k	3.31	$-37k

Recommended calendar rule — session VWAP vs closing-print execution



Both curves use the identical signal and the identical 0.65 CRC round-trip slippage; only the fill reference differs. They track each other closely and the VWAP (realistic) curve carries the higher Sharpe — the calendar edge is an economic flow, not an artifact of marking to the close.

F2. Secondary strategies at VWAP (kept for completeness)

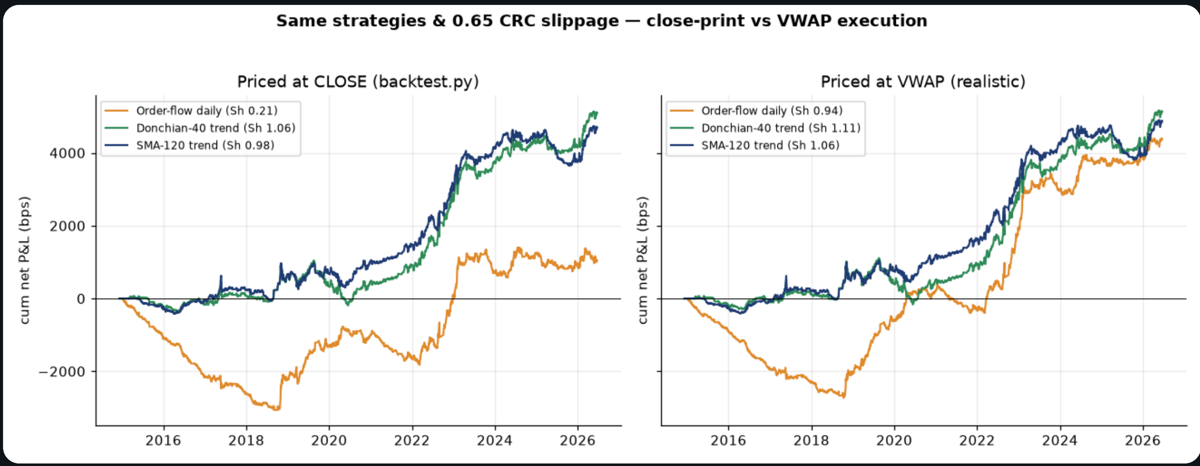
Why they're here

These trend / order-flow books are **not** the recommendation — they are shown only to demonstrate that the VWAP-execution finding generalises beyond the calendar rule. On the realistic VWAP basis the noisy daily order-flow book improves most (its turnover is where a single closing tick hurts), while the slow trend books barely move. If you only care about the recommended strategy, F1 above is the whole story.

SECONDARY STRATEGY (NET 0.65 CRC RT, VWAP-TO-VWAP)	CLOSE SHARPE	VWAP SHARPE	VWAP BPS/YR	TRADES/YR
Order-flow daily	0.21	0.94	380	87.3
Donchian-40 trend	1.06	1.11	445	2.3
SMA-120 trend	0.98	1.06	422	6.4

Close vs VWAP chart

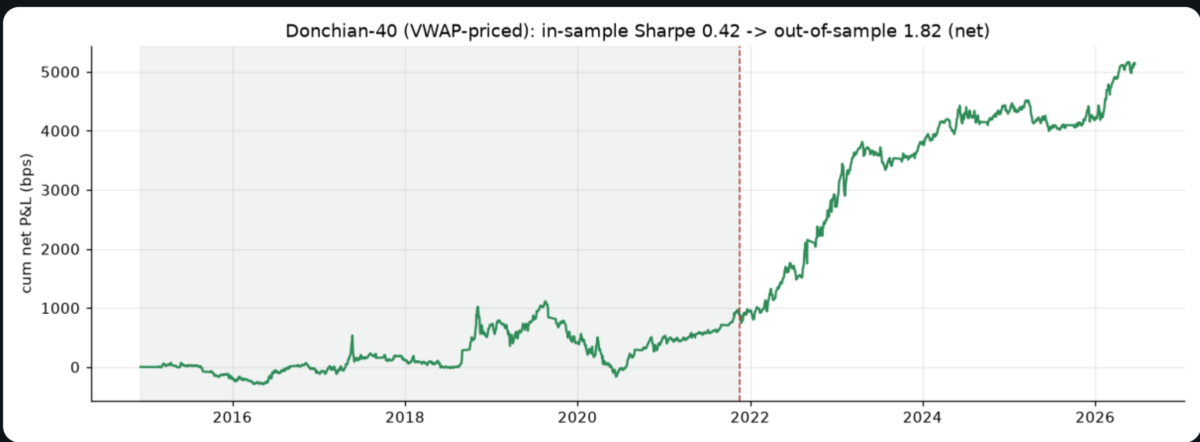
Secondary books & slippage — close-print vs VWAP execution



Left priced at the close, right at the VWAP. The trend books (Donchian-40 1.06→1.11, SMA-120 0.98→1.06) barely move; the daily order-flow book improves most (0.21→0.94).

Out-of-sample

Donchian-40 (VWAP-priced): in-sample vs out-of-sample net Sharpe



Held out 60/40, the VWAP-priced Donchian-40 keeps working — OOS net Sharpe 1.82 (≥ IS 0.42).

PART G · VERDICT

The recommendation is the calendar + slow-vol rule. Short USD into Costa Rica's mid-month IVA / payroll deadline (within 6 business days of it), long the rest of the month, at **half size whenever a slow 20-day volume regime disagrees**. Full-sample $95k/yr at Sharpe 3.31, ~18 trades/yr, drawdown $-37k — and, the number that matters, **out-of-sample Sharpe 3.55** ($123k/yr on the 40% held out). Selected in-sample, confirmed out-of-sample.

The slow-vol trim is what makes it trade-ready. Ranked on in-sample Sharpe, both slow-vol variants HOLD or IMPROVE out of sample while the two strong flat-sized entries DECAY (real-calendar 3.14 → 2.79, fixed-window 3.19 → 2.67). The trim cuts the fat losing trades: out-of-sample per-trade skew turns **positive (+0.74)** and the worst trade halves ($-50k → $-18k). That tail control, not the headline P&L, is why this is the pick.

A dynamic exit is an optional second tail control (A5 / A5b). The newest exit-engine run (`exit_lab.py`, 231-session basis) settles the design: on the calendar entry, a **trailing 30 bps + hard stop 40 bps** overlay lifts the in-sample Sharpe from 3.14 to **3.7** and the out-of-sample from 2.79 to **3.92**, cutting the worst drawdown $-48k → $-26k. But a paired test (p = 0.7292) shows total P&L is unchanged — it is a **risk** improvement, a smoother path to size up, not more money. **A take-profit actively hurts** and is not recommended. It is an alternative route to the same tail control as the slow-vol trim, not a required stack.

Ranked on one honest basis (Part R), the calendar family dominates. Re-priced identically on the real 231 sessions/yr, the top three seats are the calendar entry plus an exit overlay (recommended trail 30 + floor 40, IS Sharpe 3.7 → OOS 3.92), the flow rules sit at the bottom with $-397k-class drawdowns, and the mirror-image benchmarks confirm the accounting. It is the one edge that works in both the pegged and the float regimes.

Your volume thesis is correct but must be traded gently. Low volume → USD up is real and monotonic, but a daily flip pays much of it to slippage even VWAP-priced. Its value here is exactly as the *slow filter* that sizes the calendar trade — that is what turns the plain calendar rule into the recommended one — and, conviction-sized, lifts the combined/ML book to Sharpe 2.03–3.84.

The BCCR's hand is now visible, and it sharpens the risk controls (Part I). Official flow (BCCR + public sector) is a median 52% of MONEX volume and peaks at the deadline, exactly where the colón turns — it *names* the reversion the exit overlay harvests. The tradeable use is the **reserves**: trimming long-USD size when reserves rise lifts out-of-sample Sharpe 3.92 → 4.09 and cuts drawdown $-26k → $-22k, beating a blind long-trim out-of-sample (p = 0.001). A risk improvement, not new alpha — and strictly causal.

The mechanism is economic, not technical: exporter USD supply (volume) and the payment/tax cycle (quincena). That is why it persists across regimes — and why it would weaken if the BCCR re-pegged or intervened heavily, the dominant risk.

Priced realistically, the edge survives. Everything here is marked at the session VWAP, not the closing print. For the recommended rule that is a slight *improvement* (3.2 → 3.31 Sharpe), so the result is not an artifact of marking to the close.

Dollar & pricing convention. $1,000,000 USD notional per full position; 1 bp of next-session move = $100; slippage 0.325 CRC/side (~$680) charged on every position change. **All P&L is priced VWAP-to-VWAP** — a position decided from information through session *t* earns the next session's VWAP move, and the slippage is booked against the VWAP (the realistic desk fill), net, gross of financing/borrow and market impact, non-compounded (reset to $1M each day). The ML row is conviction-sized so its notional varies up to $1M; all model numbers are walk-forward out-of-sample. **Selection discipline:** variants are ranked on the **in-sample** Sharpe (first 60%, 2014-12-08 → 2021-11-17); every headline result, histogram and per-trade statistic for the recommended rule is its **out-of-sample** realisation on the held-out 40% (2021-11-18 → 2026-06-19). Nothing is re-fit on the test window. In-sample / out-of-sample = chronological 60/40 split at 2021-11-18. Reproducible: `parse_monex → parse_bccr → analyze → … → rank_strategies → exit_lab → intervention → build_report`. Every part is annualised on the venue's real 231 sessions/yr, priced VWAP-to-VWAP net of the 0.65 CRC round trip, from a single basis (`src/basis.py`) — so the per-module tables (A–F) and the unified ranking (R) agree by construction. Part I adds two BCCR exports (FX intervention CodCuadro 1587, net reserves CodCuadro 8) via `parse_bccr.py`.